Advancements in Semisupervised Remote Sensing Segmentation Using Adaptive Patch Augmentation and Class Ranking Weight

Panpan Zhu, Jiazheng Yan, Enxi Wang, Yunlong Hou, Jiale Liu, Ligang Zhou, and Xiaobo Luo

***Abstract*—Traditional deep semantic segmentation methods rely heavily on large-scale, densely labeled data, which is costly and time-consuming to obtain. Semisupervised frameworks have emerged as a promising alternative, reducing the dependency on pixel-level annotations while enhancing segmentation performance. However, these methods still face challenges in data augmentation and class imbalance. For instance, traditional augmentation techniques such as CutMix are limited by their reliance on single random local contexts, which restricts perturbation strength and hinders model generalization. Moreover, the long-tailed class distribution in remote sensing (RS) images is often overlooked, leading to pseudo-labels that are biased toward majority classes, further exacerbating class imbalance. To address these challenges, we propose a novel semisupervised framework for RS semantic segmentation, named APRW, which incorporates adaptive patch augmentation (APA) and class rank weighting. The APA module dynamically generates adaptive patch masks (APMs) by comparing the predictions of weakly and strongly augmented inputs. These masks focus on high-confidence regions while preserving diverse boundary structures, effectively expanding the perturbation space and enhancing the model's adaptability to varied inputs. The class rank weighting module maintains memory banks for both labeled and unlabeled data, dynamically calculates class weights based on the relative ranking of pixel confidences, and adaptively fuses these weights. This design mitigates biases caused by long-tailed distributions, improves the segmentation of rare classes, and enhances overall segmentation accuracy. Extensive experiments demonstrate that our method consistently outperforms existing approaches across multiple RS datasets, including DFC22, iSAID, MER, MSL, Vaihingen, and GID-15, showcasing superior generalization and higher segmentation precision. Implementation is released at https://github.com/135az/APRW**

***Index Terms*—Class imbalance, class ranking weight, data enhancement, remote sensing (RS), semantic segmentation, semisupervised learning (SSL).**

I. Introduction

THE rapid development of remote sensing (RS) technology has significantly improved the quantity and quality of RS images [1], [2]. Today, many RS dataset feature spatial resolutions at submeter levels, greatly enhancing the information captured in each image. Semantic segmentation is a critical task in RS image processing, enabling accurate pixel classification to extract rich geographic and semantic information details. Due to its relevance in essential areas such as ecological protection and urban planning, semantic segmentation has gradually become a focal point of research in academia and industry. In recent years, deep learning technology [3], [4] has emerged as the leading approach for RS semantic segmentation. These methods, known for their exceptional feature extraction and generalization capabilities, have shown remarkable performance in segmentation tasks. However, they typically rely on fully supervised learning, necessitating pixel-level annotations for each RS image, which is time-consuming and costly [5], [6]. This dependence on high-quality annotated data has severely restricted the practical application of these methods.

To tackle the challenge of limited labeled data, semisupervised learning (SSL) has been developed, which utilizes unlabeled samples to improve model performance and has become a key area of research [7], [8]. Initially applied to image classification, SSL includes techniques such as self-training (ST) [9], co-training [10], dictionary learning [11], and label propagation [12]. Current SSL methods generally adopt two primary strategies: ST and consistency regularization (CR). ST involves selecting high-confidence pseudo-labels for retraining but can lead to overfitting, particularly when data diversity is lacking [13], [14]. By contrast, CR enforces prediction consistency despite various perturbations, which include changes in input, features, and the network itself [7], [8], [15]. Although strong data augmentations (SDAs) [16], [17], [18] can improve robustness, excessive noisy supervision may hinder training, particularly in the early stages when predictions on unlabeled data are still unreliable. Benefiting from the success of SSL in image classification, numerous studies have integrated its concepts into semisupervised semantic segmentation for RS images. ST-based methods focus on generating high-quality, diverse pseudo-labels for retraining [19], [20], [21], [22]. For instance, AGCV [23] employs alternating dual-decoder guidance and cross-view learning to address weight coupling, while DAST [24] mitigates class imbalance and distribution disparities via adaptive pseudo-label sampling and dynamic thresholding. Concurrently, CR methods [25], [26], [27] improve segmentation performance by utilizing various data augmentation techniques. For example, BCF [28]

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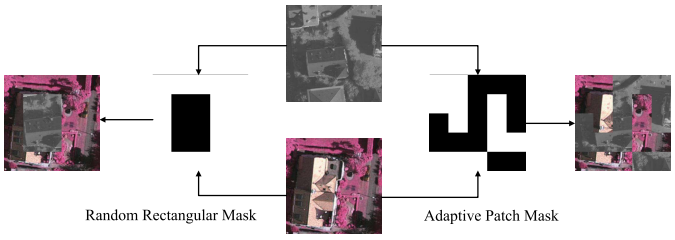


Fig. 1. (Left) Original CutMix and (right) our proposed adaptive patch CutMix using the random rectangular mask and the APM, respectively.

mitigates learning and cognitive biases via a feature consistency module, multidecoders with weakly orthogonal weights, and symmetrical knowledge transfer for robust pseudo-labels. U-MCL [27] uses patch-wise uncertainty maps for adaptive denoising and context-aware masked consistency. These methods collectively enhance segmentation performance. Unlike ST, CR does not depend on the quality of pseudo-labels; instead, it employs consistency constraints to exploit unlabeled data, offering enhanced robustness, particularly during early training and in low-confidence scenarios, thereby making it more practical for real-world applications.

The core challenge in semisupervised semantic segmentation is to make full use of the limited labeled data alongside a vast amount of unlabeled data. However, current consistency-based methods can still be improved in terms of leveraging the potential of RS images. Recent advances in semisupervised segmentation heavily rely on data augmentation, with CutMix [16] widely adopted for its strong perturbations and effectiveness in boosting performance [25], [26], [29], [30], [31]. As shown in Fig. 1, the original CutMix enhances segmentation by introducing diverse context through random patch mixing between unlabeled images, but its reliance on randomly generated rectangular masks (RRMs) limits spatial diversity and impairs precise attention to key target areas. In addition, real-world RS semantic segmentation often exhibits a long-tailed distribution, where common classes dominate and rare ones are underrepresented. This imbalance worsens in semisupervised settings, as pseudo-labels are biased toward majority classes, reducing performance on minority categories. Furthermore, existing methods for labeled images mainly rely on supervised training alone, which fails to fully exploit the potential of labeled data and, in turn, restricts the enhancement of segmentation performance.

To address these challenges, we propose a novel semisupervised segmentation method (APRW), which integrates adaptive patch augmentation (APA) and class ranking weighting (CRW). The APA module overcomes the limitations of random rectangular masks by dynamically adjusting the size and shape of the cropping box based on confidence metrics. Specifically, it computes confidence matrices from weakly and strongly augmented images, divides these matrices into small blocks, and calculates the average confidence for each block. When these blocks are compared and reconstructed, the adaptive patch mask (APM) is generated, preserving high-confidence regions while introducing diverse boundary shapes. To further enhance sample diversity, we apply two different strong augmentations to the same weakly augmented image, resulting in two sets of strongly augmented samples. These samples are then combined using APM to create mixed samples that are rich in feature information, thereby expanding the perturbation space and improving the model's ability to learn complex patterns. Concurrently, the CRW module addresses class imbalance by dynamically assigning weights to each pixel based on its confidence ranking within the predicted class. Unlike conventional methods that rely solely on unlabeled data, CRW incorporates labeled data into the weight computation to maximize its utility. By adaptively combining the weights from both data sources, this module effectively balances their effects, strengthens the model's focus on rare classes, and alleviates bias caused by a long-tailed distribution.

In summary, the main contributions of this work are as follows.

1) We present a semisupervised RS image semantic segmentation method that integrates APA and CRW modules. This approach fully leverages both labeled and unlabeled data while addressing class imbalance, resulting in notable improvements in segmentation accuracy and generalization.
2) The APA module is developed to select high-confidence regions for generating APMs, thereby expanding the perturbation space and enhancing segmentation performance, particularly in complex scenes.
3) The CRW module is designed to dynamically compute class weights by incorporating labeled and unlabeled predictions, with final weights generated through adaptive fusion. This effectively mitigates class imbalance, strengthens attention to rare classes, and improves segmentation robustness.

II. Related Work

A. Semisupervised Image Segmentation

In semisupervised semantic segmentation, several effective techniques have emerged, where the most notable are adversarial methods, ST, consistency learning, and contrastive learning. Adversarial methods, inspired by GANS [32], enhance improve the model's ability to leverage unlabeled data through a generator–discriminator framework. The generator produces pseudo-labels or samples, whereas the discriminator assesses their quality, gradually guiding the model to utilize unlabeled data more effectively. Techniques such as GCT [33] and S4GAN [34] have achieved strong segmentation performance through adversarial training. However, their sensitivity to hyperparameters and potential noise in pseudo-labels limits their effectiveness in complex segmentation tasks. To overcome these challenges, ST methods have gained popularity. These methods generate pseudo-labels from a model trained on labeled data and incorporate them into subsequent training to enhance generalization. ST++ [35] exemplifies this approach by progressively refining pseudo-labels, significantly boosting segmentation performance. Building on ST, consistency learning enhances semisupervised segmentation by enforcing prediction consistency across augmented inputs, thereby reducing model sensitivity to perturbations.

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CorrMatch [36] and SPEED [37] enhance pseudo-label quality and consistency via correlation maps, entropy-based augmentation, and self-distillation without relying on complex training designs. More recently, contrastive learning has broadened the scope of semisupervised segmentation by optimizing feature relationships, improving intraclass consistency, and enhancing interclass separation. Methods like RankMatch [20], [38] apply this technique to more effectively utilize unlabeled data, thereby improving segmentation performance.

These diverse technological approaches have not only driven significant advancements in the performance and generalization capabilities of semisupervised semantic segmentation methods, but they have also opened up extensive opportunities for practical applications, such as RS image processing.

B. Semisupervised RS Image Segmentation

For the task of semisupervised semantic segmentation of RS images, various methods have been proposed, chiefly categorized into ST and consistency learning strategies. In ST methods, the two-stage ST framework proposed in [19] leverages cross-entropy (CE) and mse losses for initial training, followed by pseudo-label generation. However, it risks noise accumulation from erroneous pseudo-labels. LSST [21] improves pseudo-label reliability through a linear sampling strategy, which emphasizes confident predictions but may overlook hard samples with potential informative value. To address class imbalance, Huang et al. [22] introduced a decoupled weighted learning framework, incorporating ranking-based weights to suppress pseudo-label noise in minority classes. However, its performance is still constrained by the initial pseudo-label quality. Consistency learning methods include RanPaste [25], ICNet [39], WSCL [26], and MCSS [31], among others. RanPaste [25] enhances learning with a novel random masking-based augmentation that fuses labeled and unlabeled data, but its reliance on handcrafted thresholds may limit generalization. WSCL [26] proposes sparse dual-view generation and an adaptive weighting mechanism based on entropy maps, improving the model's focus on uncertain regions, though at the cost of increased complexity. MCSS [31] enhances feature discriminability through multidisturbance consistency learning and a class prototype memory, but its dependence on additional memory structures and intricate alignment operations may limit scalability in resource-constrained settings. Unlike augmentation-based methods, ICNet [39] leverages a dual teacher–student framework with iterative updates to enhance generalization across datasets, but its iterative updates may slow convergence for large-scale RS datasets.

These methods have advanced semisupervised RS segmentation by refining pseudo-label quality, introducing new augmentation strategies, and improving consistency learning. However, they have not fully explored the potential of RS images, especially in terms of fully leveraging image-specific characteristics and addressing class imbalance and rare class performance. To address these shortcomings, we have improved existing techniques and proposed a novel semisupervised semantic segmentation method aimed at more efficiently utilizing limited labeled data and abundant unlabeled data.

III. Method

As illustrated in Fig. 2, the proposed end-to-end semisupervised RS image segmentation method (APRW) integrates two key modules, namely APA and CRW, effectively enhancing the utilization efficiency of both labeled and unlabeled data, as well as improving the segmentation performance of the model. Specifically, the APA module generates diverse mixed samples by dynamically focusing on high-confidence regions within unlabeled samples, thereby enhancing the model's adaptability to varied inputs. On the other hand, the CRW module dynamically adjusts class weights, guiding the model to prioritize the learning of features from challenging classes, further boosting the overall segmentation performance. The implementation details of the APA and CRW modules will be thoroughly described in Sections III-B and III-C, respectively.

A. Problem Definition

Given a labeled dataset $D_l = \{(x_i^l, y_i^l)\}_{i=1}^{N_l}$ and an unlabeled dataset $D_u = \{x_i^u\}_{i=1}^{N_u}$, where $x_i^l \in \mathbb{R}^{H\times W\times 3}$ represents the ith labeled input image, $y_i^l \in \mathbb{R}^{H\times W\times C}$ is its corresponding pixel-level ground truth label, C is the number of classes, x_i^u is the ith unlabeled input image, and N_l and N_u denote the number of labeled and unlabeled images, respectively. Following LSST [21] and WSCL [26], we resample the labeled dataset D_l such that $N_l \approx N_u$. The goal of semisupervised RS image semantic segmentation is to train a network f_θ with parameters θ to perform pixel-level classification. The overall objective function can be expressed as follows:

$$L = L_s + \lambda L_u \tag{1}$$

where λ is a hyperparameters used to balance the supervised loss L_s and the unsupervised loss L_u. In this study, both the supervised loss L_s and the unsupervised loss L_u adopt the standard CE loss.

For labeled images, L_s computes the CE loss between the model's prediction $f_\theta(x_i^l)$ and the ground truth label y_i^l

$$L_s = \frac{1}{N_l}\sum_{i=1}^{N_l} L_{\text{ce}}\left(y_i^l, f_\theta\left(x_i^l\right)\right) \tag{2}$$

where $L_{\text{ce}}(\cdot,\cdot)$ represents the CE loss.

As for unlabeled data, following the CR framework from weak to strong in SSL, each unlabeled image x_i^u undergoes two sets of augmentation operations before being fed into the network: weak augmentation $A^w(\cdot)$ (e.g., scaling and cropping) and strong augmentation $A^s(\cdot)$ (e.g., color jittering and random grayscale), generating versions $u_i^w = A^w(x_i^u)$ and $u_i^s = A^s(A^w(x_i^u))$ with different perturbation intensities. Table I lists the specific contents of the two augmentation operations. It is worth noting that in the strong perturbation operation $A^s(\cdot)$, we replace the original CutMix with the proposed adaptive patch-based CutMix perturbation, the details of which will be elaborated in Section III-B. L_u computes the CE loss between different predictions derived from the same image, obtained through weak and strong augmentations, respectively, and

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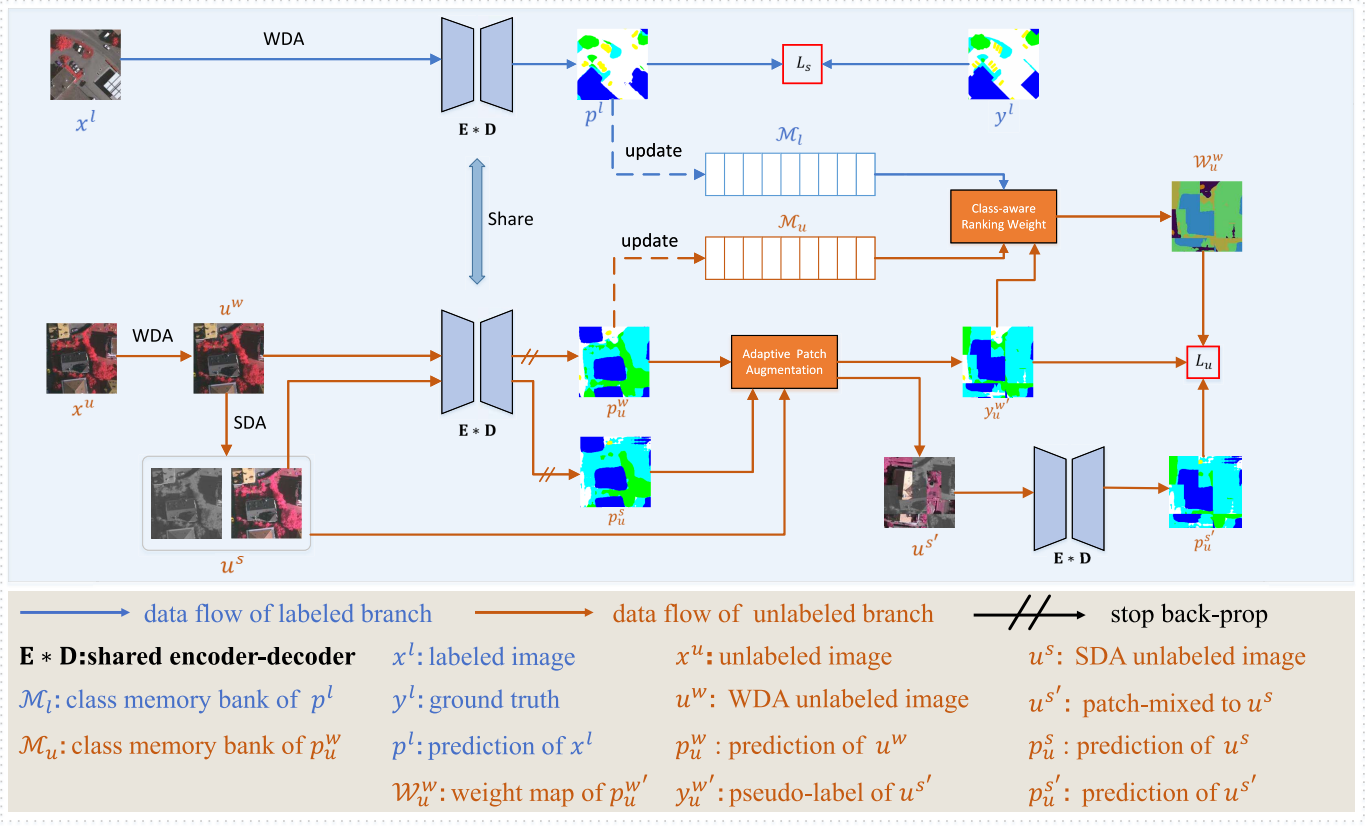


Fig. 2. Proposed APRW framework consists of a shared encoder–decoder architecture $E * D$ and two modules: adaptive patch data augmentation and CRW. The APA module dynamically generates APM by comparing the weak enhancement prediction p_u^w with the strong enhancement prediction p_u^s of the same image and applies it to u^s to generate the enhanced sample $u^{s'}$. The CRW module calculates the weight W_u^w of the enhanced sample prediction $y_u^{w'}$ based on the sampling results in the class memory library, which is used to calculate the unsupervised loss L_u, thereby alleviating the class imbalance problem and improving the performance of rare categories.

TABLE I
AUGMENTATION TECHNIQUES AND THEIR DESCRIPTIONS

Symbol	Technique	Description
A^w	Random Scale	Randomly resizing the image by [0.5, 2.0]
	Random Crop	Randomly cropping a region from the image
	Random Flip	Horizontally flipping the image with a probability of 0.5
A^s	Color Jitters	Adjusting brightness, contrast, saturation, and hue with probabilities (0.5, 0.5, 0.5, 0.25)
	Random Gray Scale	Converting images to grayscale with a probability of 0.2
	Edge Enhance	Applying edge enhancement filter with a probability of 0.5
	Adaptive CutMix	The proposed patch-based CutMix transformation

designates the model's prediction under weak augmentation as the pseudo-label y_i^u

$$L_u = \frac{1}{N_u} \sum_{i=1}^{N_u} L_{\text{ce}} \left(y_i^u, f_\theta \left(u_i^s \right) \right) \tag{3}$$

$$y_i^u = \arg\max \left(f_\theta \left(u_i^w \right) \right). \tag{4}$$

In the following subsections, we will detail the proposed APA module and the computation process of the pixel-level consistency loss L_u based on the CRW module.

B. Adaptive Patch-Based Augmentation

CutMix [16] is a data augmentation method widely applied in SSL. Its core idea is to generate strongly perturbed augmented samples by cutting and pasting random regions of unlabeled samples, thereby improving the model's generalization ability. Specifically, CutMix uses a randomly generated binary mask $M \in \{0, 1\}^{H \times W}$ to determine the regions sampled from two images. For two unlabeled images x_1^u and x_2^u, the goal of CutMix is to perturb x_1^u by pasting a portion of the other image x_2^u, generating an augmented sample $x_1^{u'}$. The formula is as follows:

$$x_1^{u\prime} = \text{CutMix} \left(x_1^u \right) = M \odot x_1^u + (1 - M) \odot x_2^u \tag{5}$$

where $\odot$ denotes the element-wise multiplication operation, and M is a randomly generated mask matrix that determines the proportion of regions sampled from images x_1^u and x_2^u. In traditional methods, the size of the mask region in CutMix is randomly determined. Although such randomly generated masks can provide basic perturbation effects, they often fail to effectively cover key target regions and may introduce a large amount of useless or redundant information. In cases of imbalanced class distributions or high sample complexity, such masks can easily lead to instability in model training, particularly during the early stages of training or when handling difficult samples. To address the aforementioned issues and further enhance the representativeness and information

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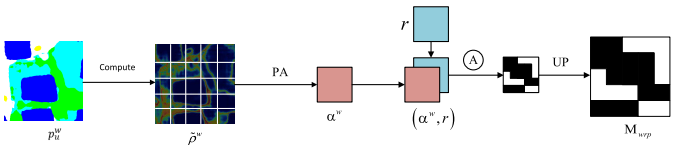


Fig. 3. Structure of WRPM. PA: patch average, Ⓐ: argmax, and UP: upsample.

diversity of strongly augmented samples, we introduce a patch-based operation and explore the impact of different patch masks on the effectiveness of CutMix augmentation. Specifically, we propose patch masks generated by comparing weakly augmented confidence matrices with random matrices [weakly random patch masks (WRPMs)] and by comparing weakly augmented with strongly augmented confidence matrices [weakly strongly patch masks (WSPMs)].

1) Methodology for Generating WRPM: The generation process of WRPM is illustrated in Fig. 3. Specifically, we first calculate the confidence matrix based on the model's prediction results. The confidence matrix measures the confidence of each predicted pixel belonging to different categories, reflecting the reliability of the model under different perturbations. The confidence metric is designed as follows:

$$\rho = \max\left(p\left(x^u\right)\right)\left(1 - \frac{-\sum p\left(x^u\right)\log\left(p\left(x^u\right)\right)}{\log C}\right) \tag{6}$$

where $p(x^u)$ represents the prediction result of the unlabeled data, and C denotes the number of categories in the dataset. For weakly augmented data u^w, we can compute its confidence matrix ρ^w. Then, ρ^w is divided into nonoverlapping patches $\tilde{\rho}^w \in \mathbb{R}^{H/n\times W/n}$, and the average confidence of each patch is calculated as follows:

$$\tilde{a}^w = \frac{n^2}{H\times W}\sum_{j=1}^{\frac{H\times W}{n^2}} \tilde{\rho}_j^w. \tag{7}$$

Simultaneously, a random matrix r with the same shape as $\tilde{a}^w$ is generated, where each element follows a uniform distribution between 0 and 1. The set $\{(\tilde{a}^w, r)\}_{m=1}^{n^2}$ is then reconstructed into a pair of patch-level average confidence maps $(\alpha^w, r) \in \mathbb{R}^{n\times n}$ based on their corresponding positions. Finally, the binary mask $\mathrm{M_{wrp}}$ can be derived as follows:

$$\mathrm{M_{wrp}} = \mathrm{Upsample}\left(\arg\max\left(\mathrm{concat}\left(\alpha^w, r\right)\right)\right) \tag{8}$$

where Upsample(·) represents upsampling the processed binary result to the original image size.

2) Methodology for Generating WSPM: For WSPM, the entire process is similar to that of WRPM, as illustrated in Fig. 4. For weakly augmented data u^w and strongly augmented data u^s obtained from the same image, we calculate their respective confidence matrices ρ^w and ρ^s using (6). Next, (ρ^w, ρ^s) is divided into nonoverlapping patches $(\tilde{\rho}^w, \tilde{\rho}^s) \in \mathbb{R}^{H/n\times W/n}$, and the average confidence of each patch is computed as follows:

$$\tilde{a}^w = \frac{n^2}{H\times W}\sum_{j=1}^{\frac{H\times W}{n^2}} \tilde{\rho}_j^w, \quad \tilde{a}^s = \frac{n^2}{H\times W}\sum_{j=1}^{\frac{H\times W}{n^2}} \tilde{\rho}_j^s. \tag{9}$$

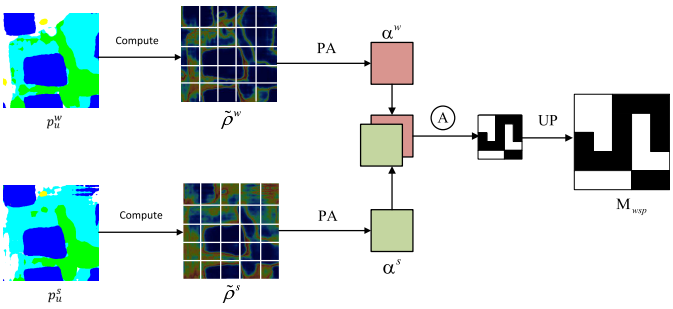


Fig. 4. Structure of WSPM. PA: patch average. Ⓐ: argmax. UP: upsample.

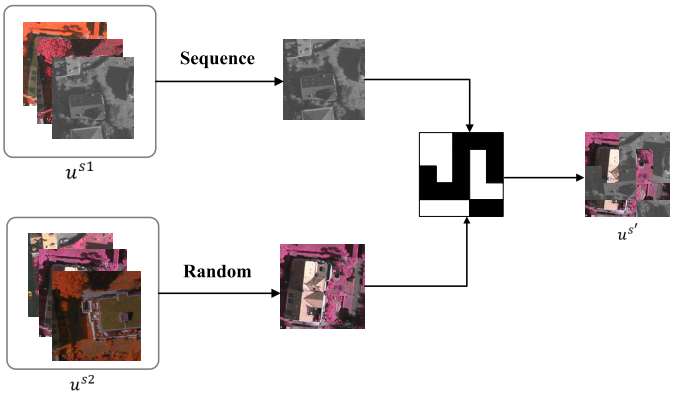


Fig. 5. Image paste method: merging two sample sets at the patch level based on APM rules to generate new training samples.

Then, the set $\{(\tilde{a}^w, \tilde{a}^s)\}_{m=1}^{n^2}$ is reconstructed into a pair of patch-level average confidence maps $(\alpha^w, \alpha^s) \in \mathbb{R}^{n\times n}$ based on their corresponding positions. Finally, the binary mask $\mathrm{M_{wsp}}$ can be derived as follows:

$$\mathrm{M_{wsp}} = \mathrm{Upsample}\left(\arg\max\left(\mathrm{concat}\left(\alpha^w, \alpha^s\right)\right)\right). \tag{10}$$

As shown in Fig. 5, to make the strongly augmented samples more representative and capable of containing rich and diverse information, we perform two different strong augmentation operations on the same weakly augmented image, generating two sets of strongly augmented samples u^{s1} and u^{s2}

$$u^{s1} = A^{s1}\left(A^w\left(x_1^u\right)\right), u^{s2} = A^{s2}\left(A^w\left(x_1^u\right)\right) \tag{11}$$

where $A^{s1}(\cdot)$ and $A^{s2}(\cdot)$ represent two distinct strong augmentation operations. APM is the adaptive patch mask generated based on the first set of strongly augmented samples u^{s1}. Therefore, for the first set of strongly augmented samples, we apply the CutMix augmentation operation sequentially, while for the second set of strongly augmented samples u^{s2}, supplementary information is generated through random selection. Subsequently, based on the corresponding rules of APM, the two sets of samples are combined at the patch level to generate new training samples $u^{s'}$

$$u^{s'} = \mathrm{APM}\odot u^{s1} + (1 - \mathrm{APM})\odot u^{s2}. \tag{12}$$

Through this APM-based CutMix perturbation method, the interference information introduced into a given image can be dynamically controlled, allowing the strength of data augmentation to be adjusted according to the training phase. This

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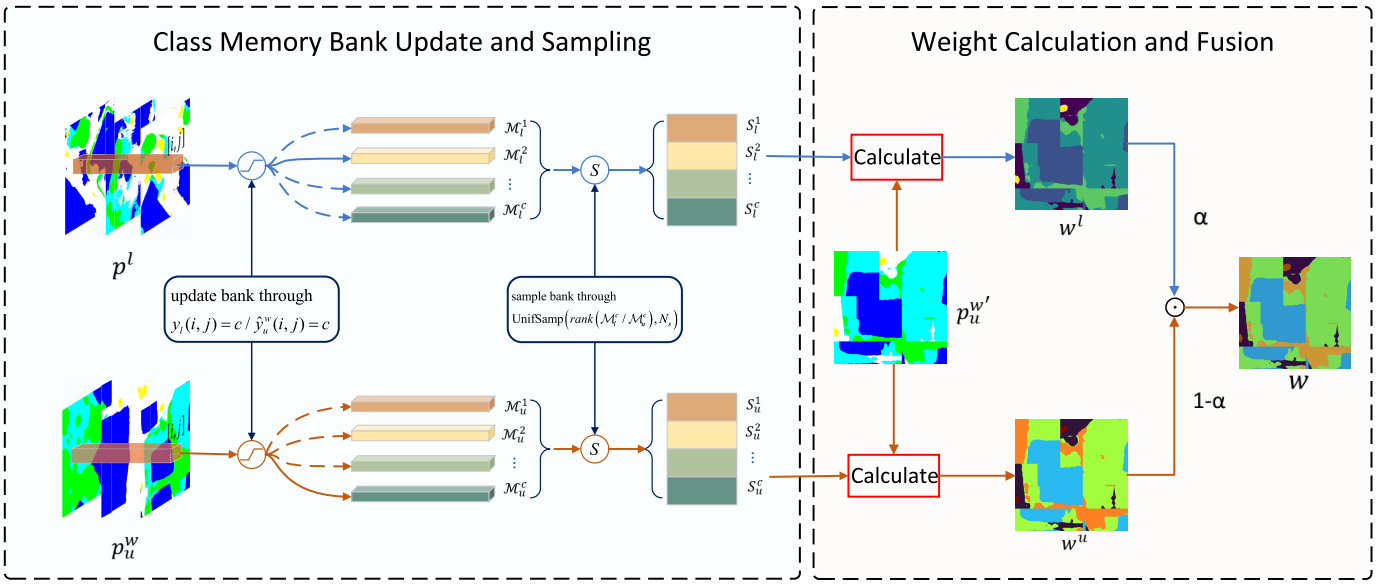


Fig. 6. Mechanism of feature update and weight fusion based on memory bank. Class-wise memory banks are dynamically updated with pixel-level predictions from labeled and pseudo-labeled data (y_l and $\hat{y}_u^w$) and uniformly sampled to streamline the computation of pixel-wise weights.

design not only captures the prediction differences between weakly and strongly augmented image versions more effectively but also guides the model to focus on high-confidence regions, thereby achieving more stable model training and further improving segmentation performance and the reliability of prediction results.

C. Class Memory Rank Weight

Although the aforementioned data augmentation methods have shown effectiveness in expanding the perturbation space and mitigating model bias, their performance remains limited when addressing the typical long-tailed class distribution problem in RS images. The scarcity of samples from rare classes makes it difficult for the model to effectively learn the features of these classes, leading to pseudo-labels that are biased toward majority classes and exacerbating the issue of class imbalance. To tackle this problem, we draw inspiration from the existing class memory bank ranking weight mechanism [22] and introduce improvements in two key aspects: first, by incorporating the prediction results of labeled data to more comprehensively integrate and utilize information from different data sources; and second, by adopting an adaptive weight fusion method to achieve a dynamic balance between the contributions of labeled and unlabeled data. These improvements effectively alleviate the class imbalance issue, and the segmentation performance of the model is significantly enhanced. Consequently, the pixel-level consistency loss function based on the CRW module can be expressed as

$$L_u = \frac{1}{\mathrm{HW}} \sum_{i=1}^{H} \sum_{j=1}^{W} w_{\mathrm{ij}} \cdot L_{\mathrm{ce}} \left(y_{\mathrm{ij}}^{u'}, f_\theta \left(u_{\mathrm{ij}}^{s'} \right) \right) \tag{13}$$

where $u_{\mathrm{ij}}^{s'}$ and $y_{\mathrm{ij}}^{u'}$ represent the training samples and pseudo-labels, respectively, after adaptive patch-based data augmentation, and $w \in \mathbb{R}^{H \times W}$ is the weight map for $y^{u'}$. In each iteration, we update the class memory module based on the prediction results from both labeled and unlabeled data to ensure that the weight calculation is based on the latest feature representations. We employ independent storage mechanisms for the weights generated from labeled and unlabeled data to facilitate subsequent weighted merging. The entire process of this module includes three parts: memory bank update and sampling, weight calculation and fusion, and adaptive weight update. Fig. 6 illustrates the first two parts.

1) Updating Memory Banks: To better exploit both labeled and unlabeled data, we maintain a class-wise memory bank

$$\mathcal{M} = \bigcup_{c=1}^{C} \left(\mathcal{M}_u^c \cup \mathcal{M}_l^c \right) \tag{14}$$

which stores pixel-level prediction probabilities for each class. Here, $\mathcal{M}_u^c$ and $\mathcal{M}_l^c$ denote the memory banks from unlabeled and labeled data for class c, respectively. For u^w, the probability map p_u^w and pseudo-label map $\hat{y}_u^w$ are resized to 64×64 to reduce storage overhead. At each pixel location (i, j), if $\hat{y}_u^w(i, j) = c$, the corresponding prediction vector $p_u^w(i, j)$ is stored in $\mathcal{M}_u^c$

$$\mathcal{M}_u^c = \mathcal{M}_u^c \cup \left\{ p_u^w(i, j) \mid \hat{y}_u^w(i, j) = c \right\}. \tag{15}$$

Similarly, for labeled images, we store $p_l(i, j)$ into $\mathcal{M}_l^c$ if the ground truth label $y_l(i, j) = c$

$$\mathcal{M}_l^c = \mathcal{M}_l^c \cup \{ p_l(i, j) \mid y_l(i, j) = c \}. \tag{16}$$

To ensure memory efficiency, each bank $\mathcal{M}_u^c$ and $\mathcal{M}_l^c$ is limited to a maximum size of $N_{\max}$. When the number of stored vectors exceeds this threshold, we use a first-in-first-out (FIFO) strategy to remove the oldest entries. This dynamic update maintains a compact and up-to-date feature representation for each class. In our experiments, $N_{\max}$ is set to 50.

2) Sampling Memory Banks: For a pixel (denoted as $p_{\mathrm{ij}}^{\mathrm{wu}}$) in the weakly augmented unlabeled prediction map, its weight is positively correlated with the ascending rank of its pseudo-label in the corresponding class memory bank, denoted as w_{ij}^u,

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where $w_{ij}^{u} \in [0, 1]$. The data in the memory bank is sorted based on prediction values, where a higher rank indicates that the prediction is closer to a high-confidence sample, thus assigning a higher weight; conversely, a lower rank corresponds to a lower weight. To reduce the computational burden, we adopt a uniform sampling strategy, selecting N_s samples from these memory banks to participate in subsequent weight calculations. For each class c, the sampled set is obtained as follows:

$$S_u^c = \text{UnifSamp}\left(\text{rank}\left(\mathcal{M}_u^c\right), N_s\right) \tag{17}$$

where rank$(\cdot)$ represents the sorting operation and UnifSamp$(\cdot, N_s)$ denotes uniformly sampling N_s samples. Similarly, for labeled data, the same principle and sampling method are applied to obtain the sample set

$$S_l^c = \text{UnifSamp}\left(\text{rank}\left(\mathcal{M}_l^c\right), N_s\right). \tag{18}$$

In this study, N_s is set to 20.

3) Calculating and Fusing Weights: To more accurately reflect the characteristics of class distribution, this study incorporates the sampling results of labeled data into the weight calculation. By integrating both pseudo-labels and labeled data predictions, the method mitigates potential biases that may arise from relying solely on pseudo-labels, leading to a more balanced weight distribution. Specifically, for each pixel in the weakly augmented unlabeled prediction map, the weight can be computed as follows:

$$w_{ij}^{u} = \frac{1}{N_s}\sum_{n=1}^{N_s} 1\left(p_{ij}^{wu} > S_u^c(n)\right), \quad c = \hat{y}_u^w(i,j) \tag{19}$$

$$w_{ij}^{l} = \frac{1}{N_s}\sum_{n=1}^{N_s} 1\left(p_{ij}^{wu} > S_l^c(n)\right), \quad c = y_l(i,j) \tag{20}$$

where $1(\cdot)$ is the indicator function and w_{ij}^{u} and w_{ij}^{l} represent the weights derived from unlabeled and labeled data, respectively. To further balance the contributions of labeled and unlabeled data, this study adopts a weighted fusion approach to compute the final weight for each pixel

$$w_{ij} = \alpha \cdot w_{ij}^{l} + (1-\alpha)\cdot w_{ij}^{u} \tag{21}$$

where the hyperparameter α is used to adjust the influence ratio of labeled and unlabeled data during the training phase. In this study, α is set to 0.1.

4) Updating Weights Adaptively: In the weight calculation process, the weight of a pixel is determined based on its ranking within the sampled set and increases linearly throughout training. However, as training progresses, the weight requirements fluctuate across different stages, necessitating dynamic adjustments over time. Motivated by curriculum learning [40], [41], this study employs a gradually increasing weight adjustment strategy shown to be more effective. To implement this, a novel projection function—drawing on the design of DWL [22]—is introduced to adaptively adjust weights according to the training phase, thereby better aligning with the model's learning dynamics

$$w_{ij} = \frac{2}{1+e^{-\frac{\Psi}{T}w_{ij}}} - 1 \tag{22}$$

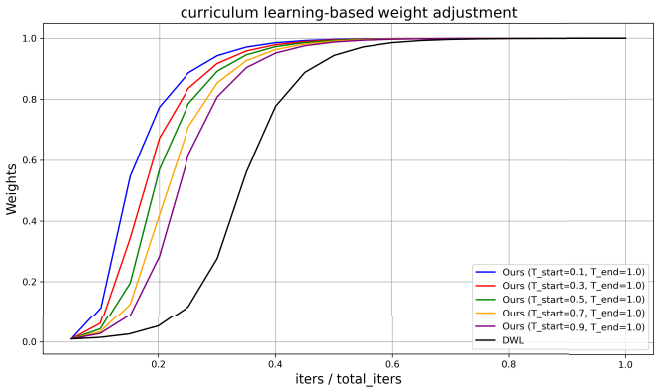


Fig. 7. Weight adjustment based on curriculum learning to adjust for more stable ST of unlabeled data at different stages as formulated by (13).

$$\Psi = 20\sqrt{\frac{\text{iters}}{\text{total_iters}}} + 2 \tag{23}$$

$$T = T_{\text{start}} + (T_{\text{end}} - T_{\text{start}})\frac{\text{iters}}{\text{total_iters}}. \tag{24}$$

As shown in Fig. 7, the key parameters T and Ψ influence the evolution of the weighting term w. This function applies a Sigmoid-based nonlinear adjustment where Ψ gradually scales the weight for smooth updates, and T controls the Sigmoid gradient, increasing linearly from T_{start} to T_{end} to enable gradual adaptation. We choose $T_{\text{start}} = 0.5$ and $T_{\text{end}} = 1.0$ as it achieves a desirable balance between training stability in the early stages and convergence efficiency. This dynamic adjustment helps prevent premature convergence and overfitting. Finally, the weight w_{ij} is used in (13) to compute the pixel-level consistency loss for unlabeled data based on class memory bank ranking weights.

IV. Experiments

In this section, we provide a detailed introduction to the various experiments conducted for the proposed APRW method, including comparative experiments and ablation analysis. First, Section IV-A describes the experimental datasets, evaluation metrics, parameter settings, and comparative models. Next, Section IV-B presents the comparative results of the APRW method under two partitioning protocols against supervised baseline methods and other state-of-the-art methods. Then, Section IV-C comprehensively validates the effectiveness of APRW through ablation analysis of its key components. Finally, in Section IV-D, a thorough evaluation of the computational load for the APRW method is conducted.

A. Experimental Setup

1) Datasets: To ensure fair and comprehensive evaluation, we adhere to the dataset protocols established in WSCL [26] and LSST [21]. Our experiments utilize six well-established benchmark datasets—DFC22 [42], iSAID [43], MER [44], MSL [44], Vaihingen, and GID-15 [45]—which collectively encompass a wide spectrum of RS challenges. These datasets differ in spatial resolution, scene complexity, and class distribution, thereby reflecting diverse and realistic

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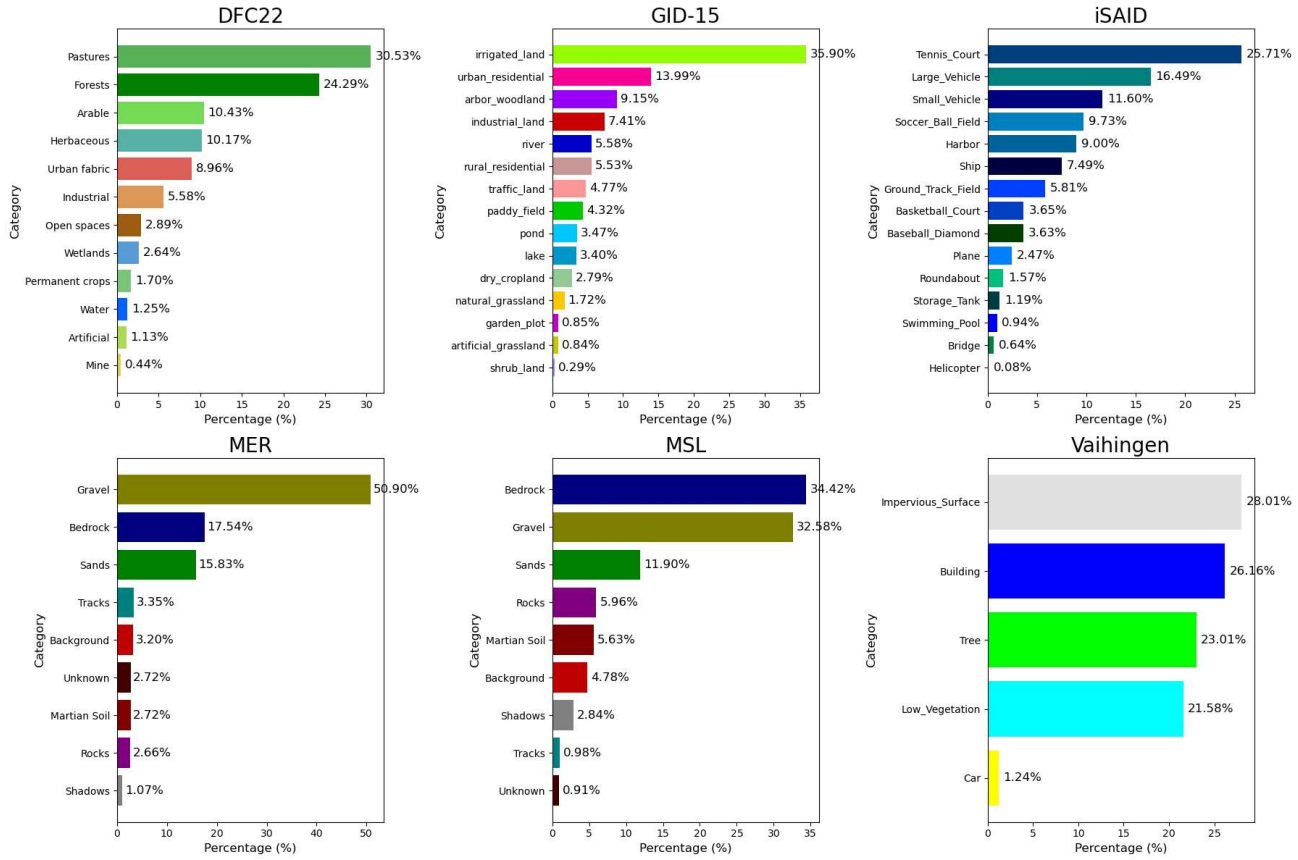


Fig. 8. Category scale distribution of each dataset: visualizing class imbalance and long-tailed characteristics in RS data.

RS environments. By partitioning each dataset with labeled ratios of 1/8 and 1/4, and treating the remainder as unlabeled data, we effectively simulate SSL scenarios with varying degrees of label scarcity. Fig. 8 visually illustrates the significant differences in class distributions across the datasets, particularly the imbalance between dominant and rare classes, highlighting the typical long-tailed distribution characteristics in RS data.

MiniFrance-DFC22 (DFC22) [42] is part of the 2022 IEEE GRSS Data Fusion Contest, containing 766 annotated images (512 × 512 pixels, 50 cm/pixels) covering diverse French landscapes. It includes 12 land cover categories: urban, industrial, mine, artificial, cropland, permanent crops, pasture, forest, herbaceous vegetation, bare land, wetlands, and water.

iSAID [43] is a large-scale aerial dataset with 2806 images and 655 451 object annotations across 15 categories. Image sizes range from 12 029 × 5014 to 455 × 387 pixels. The 15 land cover categories include: airplane, ship, storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), harbor, bridge, large vehicle (LV), small vehicle (SV), helicopter (HC), roundabout (RA), swimming pool (SP), and soccer ball field (SBF).

MER and MSL [44] originate from the Mars-Seg collection. MER consists of 4155 RGB images (560 × 500 pixels), while MSL has 1024 grayscale images (1024 × 1024 pixels). These datasets cover nine land cover categories: martian soil, sand, gravel, bedrock, rocks, tracks, shadows, background, and unknown.

Vaihingen, released by the ISPRS WG II/4 committee, represents a small village with detached and multistory buildings. It comprises 33 image tiles (1996 × 1995 to 3816 × 2550 pixels) with six land cover categories: impervious surfaces, buildings, low vegetation, trees, vehicles, and clutter.

GID-15 [45] includes ten high-resolution images (7200 × 6800 pixels) from the Gaofen-2 satellite, covering 15 categories: industrial land (InL), urban residential (UR), rural residential (RR), transportation land (TL), paddy field (PF), irrigated land (IrL), dry cropland (DC), garden plot (GP), arbor woodland (AW), shrubland (SL), natural grassland (NG), artificial grassland (AG), river, lake, and pond.

2) Evaluation Metrics: We adopt the mean intersection over union (mIoU) as the metric to evaluate segmentation performance. IoU is defined as the ratio of the intersection area between the predicted region and the ground truth region to the union area of the two. The mIoU is obtained by averaging the IoU values across all classes, and its calculation formula is as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{j=1}^{C}\frac{\text{TP}_j}{\text{TP}_j + \text{FP}_j + \text{FN}_j} \tag{25}$$

where C is the number of classes and TP_j, FP_j, and FN_j represent the number of true positives, false positives, and false negatives for the jth class, respectively.

3) Implementation Details: We use a ResNet101 [46] encoder pretrained on ImageNet and a DeepLabv2 [47]

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TABLE II
PER-CLASS IOU ON THE DFC22 TEST SET. THE BEST RESULTS ARE IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES (%)

Labeled	Method	Urban	Industrial	Mine	Artificial	Arable	Permanent	Pastures	Forests	Herbaceous	Open	Wetlands	Water	mIoU
1/8(76)	SupOnly	57.76	31.71	2.11	4.66	8.71	47.16	51.82	62.82	35.28	8.30	0.96	4.68	26.33
	UCCL	59.15	37.69	0.20	10.66	7.76	54.94	**55.57**	64.90	39.09	25.24	16.06	22.82	32.84
	AugSeg	60.31	**41.93**	6.24	12.90	9.38	62.25	53.99	67.68	41.35	18.09	13.50	22.13	34.15
	ICNet	48.83	28.02	3.51	11.58	13.32	27.78	35.26	59.15	34.62	0.98	11.59	10.35	23.75
	RanPaste	53.16	16.58	5.53	12.90	4.44	36.91	49.50	54.14	20.18	12.45	8.87	11.67	23.86
	LSST	62.49	37.59	0.80	17.34	9.82	**62.74**	55.50	64.12	36.28	15.50	1.70	9.26	31.10
	WSCL	61.08	40.55	6.52	17.36	**13.64**	62.05	50.83	69.72	**42.60**	37.48	13.49	**27.28**	36.88
	DWL	**62.62**	38.97	4.78	15.12	8.22	60.32	55.41	69.69	39.20	31.90	36.34	13.97	36.38
	Ours	60.19	41.06	**7.31**	**18.28**	13.40	61.90	52.18	**69.86**	35.34	**50.04**	**46.85**	22.01	**39.87**
1/4(153)	SupOnly	58.35	35.98	0.02	0.85	5.63	56.15	55.78	66.36	35.36	15.17	54.93	15.57	33.35
	UCCL	59.28	33.96	nan	4.01	8.04	55.05	54.97	69.14	37.59	9.07	56.58	19.43	33.93
	AugSeg	61.75	41.03	nan	1.04	6.61	55.72	55.13	69.04	41.06	45.83	55.29	20.99	37.79
	ICNet	54.02	35.23	nan	0.18	9.89	8.68	42.97	50.72	36.80	7.10	57.64	1.97	25.43
	RanPaste	50.29	28.76	nan	0.85	7.72	45.04	53.79	57.99	30.81	2.41	44.80	10.61	27.76
	LSST	58.65	**41.28**	0.07	5.32	8.46	61.35	54.05	69.04	40.29	29.74	52.81	25.07	37.25
	WSCL	58.96	39.34	nan	4.37	4.62	61.23	56.28	70.13	41.84	38.01	**60.57**	**31.10**	38.87
	DWL	63.96	38.64	nan	3.85	1.82	58.88	58.58	70.93	**42.05**	39.68	53.59	27.53	38.29
	Ours	**64.24**	41.26	**8.60**	**16.28**	**8.71**	**62.94**	**60.36**	**71.50**	41.61	**48.41**	55.20	30.64	**42.48**

decoder as the segmentation network for comparison with both supervised and state-of-the-art semisupervised semantic segmentation methods. The proposed method is implemented in PyTorch, and all experiments are conducted on one NVIDIA GTX 4090D GPU. Training is performed on six datasets with a batch size of 8, consisting of four labeled and four unlabeled images, and no warm-up stage is applied. The backbone network is optimized using stochastic gradient descent (SGD) with an initial learning rate of 0.001, a weight decay of 1×10^{-4}, and a momentum of 0.9. A polynomial learning rate scheduler dynamically adjusts the learning rate during training, following the formula $l_r = \text{base}_{l_r} \times (1 - \text{iter}/\text{total_iters})^{0.9}$. A series of data augmentation strategies is applied according to the policies and ranges listed in Table I. All models are trained for 100 epochs using uniformly cropped 321 $\times$ 321 input images across all datasets.

4) Comparative Methods: We compare our method with supervised methods and recent state-of-the-art semisupervised semantic segmentation methods, including UCCL [29], AugSeg [30], ICNet [39], RanPaste [25], LSST [21], WSCL [26], and DWL [22].

1) *SupOnly:* This supervised baseline uses a single network with ResNet-101 as the encoder and DeepLabV2 as the decoder. Only labeled samples are used during training.
2) *UCCL [29]:* It introduces semantic backpropagation for utilizing uncertain pixels and a class-aware module to ensure class-level consistency across augmented views.
3) *AugSeg [30]:* It investigates effects of various SDAs, constructs an intensity-based augmentation pool, and introduces adaptive CutMix to inject label information.
4) *ICNet [39]:* A semisupervised segmentation method based on an iterative contrastive network that leverages a teacher network to improve pseudo-label quality and model diversity through iterative training.
5) *RanPaste [25]:* It combines labeled and unlabeled data using random masks, applies consistency constraints, and sets separate confidence thresholds to reduce noise impact.
6) *LSST [21]:* It proposes an ST framework employing linear sampling to select reliable pseudo-labels for training.
7) *WSCL [26]:* It designs a sparse dual-view cross-sample image generation technique and adopts an entropy map-based adaptive reweighting strategy.
8) *DWL [22]:* It conducts ST through decoupled learning and ranking weight modules, mitigating pseudo-label errors and class imbalance.

B. Comparison Experiments

We conducted extensive experiments on the GID-15, iSAID, DFC22, MER, MSL, and Vaihingen datasets to validate the effectiveness of the proposed framework. In these experiments, we compared APRW with state-of-the-art semantic segmentation models and reported per-class IoU and mIoU evaluation metrics for each dataset. As shown in Fig. 9, our method outperforms existing advanced approaches in multiple datasets.

1) Results on the DFC22 Dataset: In the DFC22 dataset, the performance of APRW is compared with several state-of-the-art methods. The experimental results in Table II demonstrate that, compared to the fully supervised method (SupOnly), APRW achieves significant mIoU improvements of 13.54% and 9.13% on 1/8 and 1/4 labeled data, respectively, clearly showcasing its ability to effectively utilize information from unlabeled data. When compared to other semisupervised methods, our approach surpasses the second-best methods, WSCL and DWL, by 2.99% and 3.49% in mIoU on 1/8 labeled data, and this advantage increases to 3.61% and 4.19% on 1/4 labeled data, emphasizing its effectiveness in complex RS scenarios. Furthermore, our method excels in detecting rare classes. For instance, on 1/8 labeled data, the IoU for the Wetlands class reaches 46.85%, substantially outperforming WSCL's 13.49% and DWL's 36.34%. On 1/4 labeled data, the IoU for the mine class achieves 8.6%, while other methods nearly fail to detect this class. Fig. 10 illustrates the visual segmentation results of various methods on the DFC22 validation set, including the supervised baseline (SupOnly), UCCL, AugSeg, ICNet, RanPaste, LSST, WSCL, DWL, and our proposed APRW method.

2) Results on the iSAID Dataset: Table III summarizes the segmentation performance on the iSAID dataset with 1/8 and 1/4 labeled data ratios. Our method achieves an

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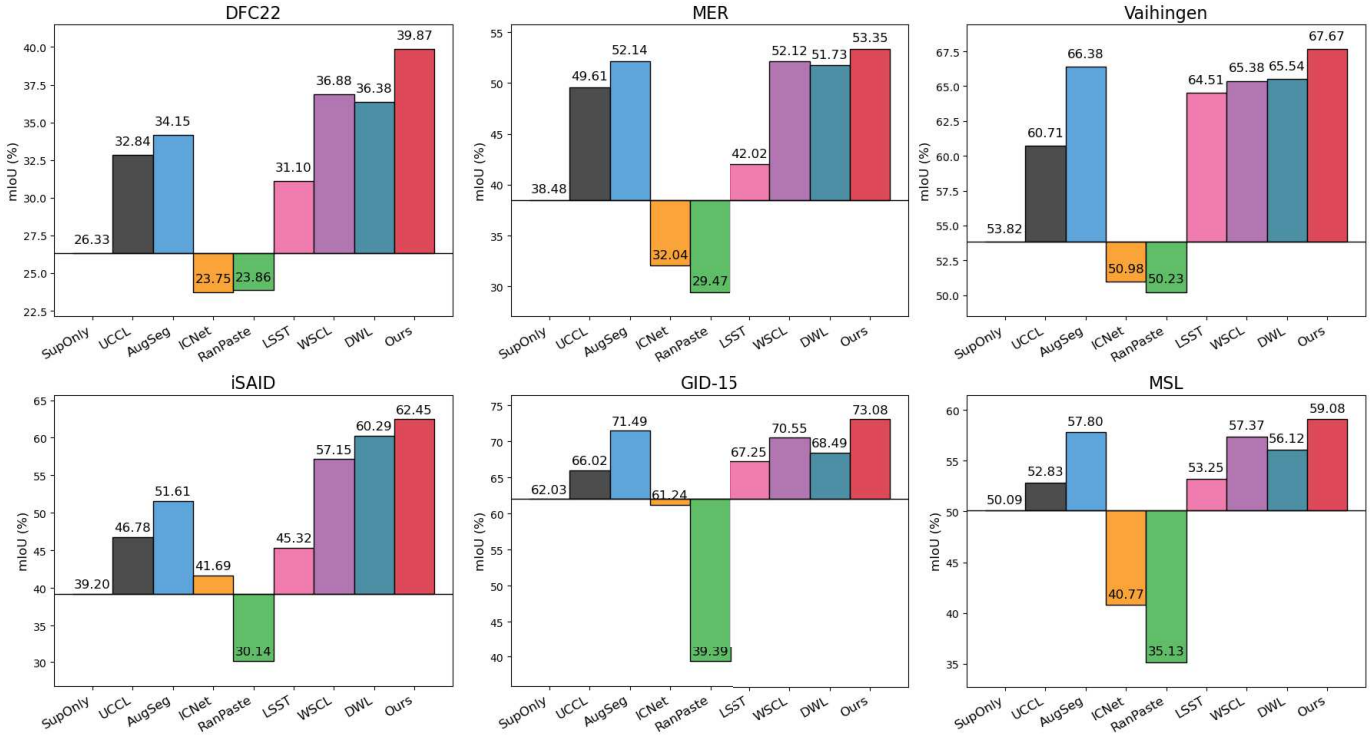


Fig. 9. Performance comparison between the proposed APRW and other advanced methods on six different RS image datasets under 1/8 partition protocol.

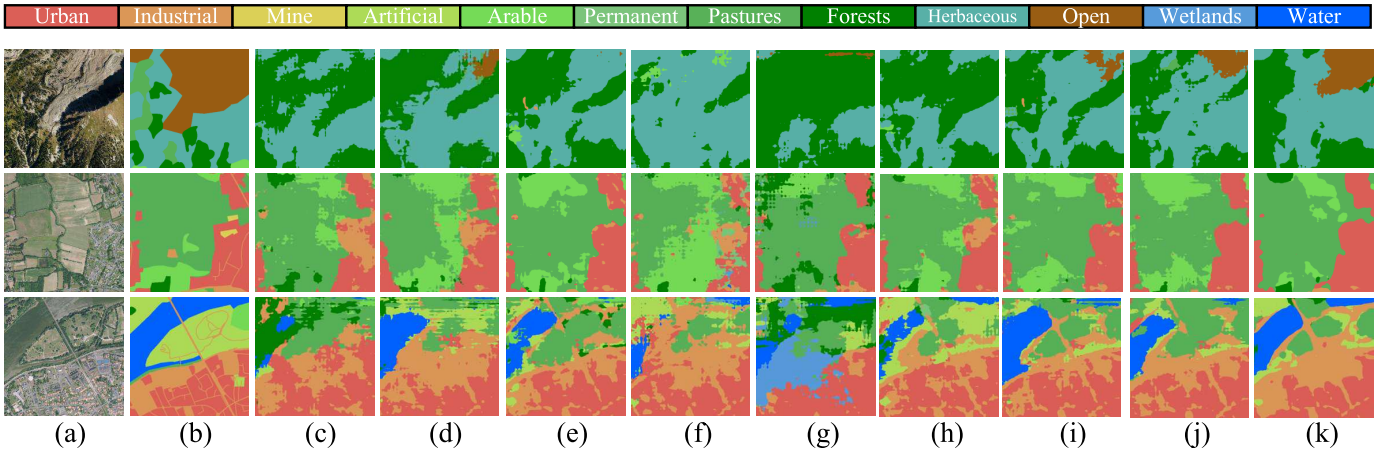


Fig. 10. Visual comparison results of the proposed method with other advanced semisupervised image segmentation methods on the DFC22 val datasets. (a) Input image. (b) Ground truth. (c) SupOnly. (d) UCCL. (e) AugSeg. (f) ICNet. (g) RanPaste. (h) LSST. (i) WSCL. (j) DWL. (k) Ours.

mIoU of 62.45% with 1/8 labeled data, which represents a 23.25% improvement over the fully supervised baseline (SupOnly, 39.20%). Compared to the second-best method, DWL (60.29%), our method improves by 2.16%. With 1/4 labeled data, our method reaches an mIoU of 71.36%, surpassing SupOnly (61.47%) by 9.89% and outperforms DWL (69.82%) with an additional 1.54% gain. Under the 1/8 labeled data protocol, the supervised baseline fails to detect the HC class, resulting in an IoU of 0 due to the absence of labeled annotations, while our method achieves 21.87% for HC, demonstrating its capability to handle unknown categories. Additionally, we achieve IoUs of 95.47% for the major category TC and 81.21% for the rare category BC, outperforming DWL's IoUs of 91.18% and 56.26%, respectively. These results illustrate that our method significantly improves overall segmentation performance with limited labeled data and excels in multiple categories.

3) Results on the MER Dataset: The results in Table IV demonstrate the strong performance of our method on the MER dataset. With 1/8 labeled data, our approach achieves an average IoU of 53.35%, reflecting a 14.87% improvement over the fully supervised baseline (SupOnly, 38.48%). Compared to other methods, our method outperforms the second-best approach, WSCL (52.12%), by 1.23%. For rare categories, such as Rocks and Tracks, our method achieves IoUs of 39.37% and 57.84%, surpassing WSCL by 5.73% and 1.89%, respectively, highlighting its superior ability to detect rare categories. With 1/4 labeled data, our method further improves

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TABLE III

PER-CLASS IOU ON THE ISAID TEST SET. THE BEST RESULTS ARE IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES (%)

Labeled	Method	Ship	ST	BD	TC	BC	GTF	Bridge	LV	SV	HC	SP	RA	SBF	Plane	Harbor	mIoU
1/8	SupOnly	37.52	4.62	75.71	81.97	12.29	21.64	2.98	71.38	45.28	nan	54.72	48.87	49.99	33.91	47.06	39.20
	UCCL	48.47	30.76	73.33	82.08	24.55	36.82	7.20	74.60	48.85	6.86	58.73	33.49	55.75	67.90	52.36	46.78
	AugSeg	39.78	1.04	74.04	91.72	67.76	49.45	1.52	78.16	49.19	5.20	**76.67**	43.24	69.19	74.89	52.36	51.61
	ICNet	36.55	35.32	50.06	73.99	28.31	14.29	3.98	70.29	43.63	3.56	61.64	60.44	47.43	54.62	41.36	41.69
	RanPaste	30.35	10.63	50.72	47.96	7.93	5.75	0.27	47.70	26.52	20.15	42.60	48.08	43.90	36.64	32.85	30.14
	LSST	39.41	0.07	81.15	80.38	14.09	31.22	nan	79.32	49.73	nan	65.88	54.82	60.90	70.75	52.04	45.32
	WSCL	41.63	**86.59**	72.76	91.27	67.58	38.28	14.85	76.19	56.51	nan	66.24	51.03	66.30	74.83	53.14	57.15
	DWL	39.67	85.36	**79.71**	91.18	56.26	46.63	**27.27**	78.11	57.40	21.31	64.44	57.89	67.94	**79.73**	51.49	60.29
	Ours	**54.32**	77.31	73.19	**95.47**	**81.21**	**62.73**	5.15	**78.79**	**56.81**	**21.87**	55.88	**81.40**	**71.52**	66.70	**54.34**	**62.45**
1/4	SupOnly	46.51	85.33	65.05	94.78	85.32	49.52	21.69	78.55	62.48	16.12	73.74	53.71	64.08	71.47	53.67	61.47
	UCCL	**54.59**	84.46	85.57	96.61	85.54	58.22	11.20	82.58	63.96	16.83	67.43	53.73	76.35	75.73	56.48	64.62
	AugSeg	44.22	86.31	88.80	97.09	88.75	65.52	27.28	81.51	67.34	25.01	**82.37**	51.10	79.09	78.57	54.51	67.83
	ICNet	38.41	86.03	49.44	94.51	78.23	52.52	12.09	78.18	54.76	29.76	75.01	68.78	60.52	66.49	47.57	59.49
	RanPaste	31.25	83.67	66.66	78.12	56.60	50.38	24.22	66.36	47.90	29.36	51.89	35.31	44.08	64.48	36.32	51.11
	LSST	48.41	82.93	57.38	96.55	78.90	59.43	23.79	78.64	61.06	22.24	71.33	**72.37**	71.65	72.04	55.02	63.45
	WSCL	46.50	87.78	86.25	96.57	88.47	66.53	**28.98**	80.85	67.10	46.09	81.97	46.90	76.47	**78.77**	55.69	68.99
	DWL	43.56	85.98	85.03	97.04	89.40	73.28	23.76	76.51	61.12	65.27	80.04	61.26	78.69	71.07	55.34	69.82
	Ours	52.45	**89.18**	**91.30**	**98.54**	**92.33**	**74.68**	28.90	**84.42**	**69.62**	**51.09**	80.66	40.95	**81.48**	76.14	**58.60**	**71.36**

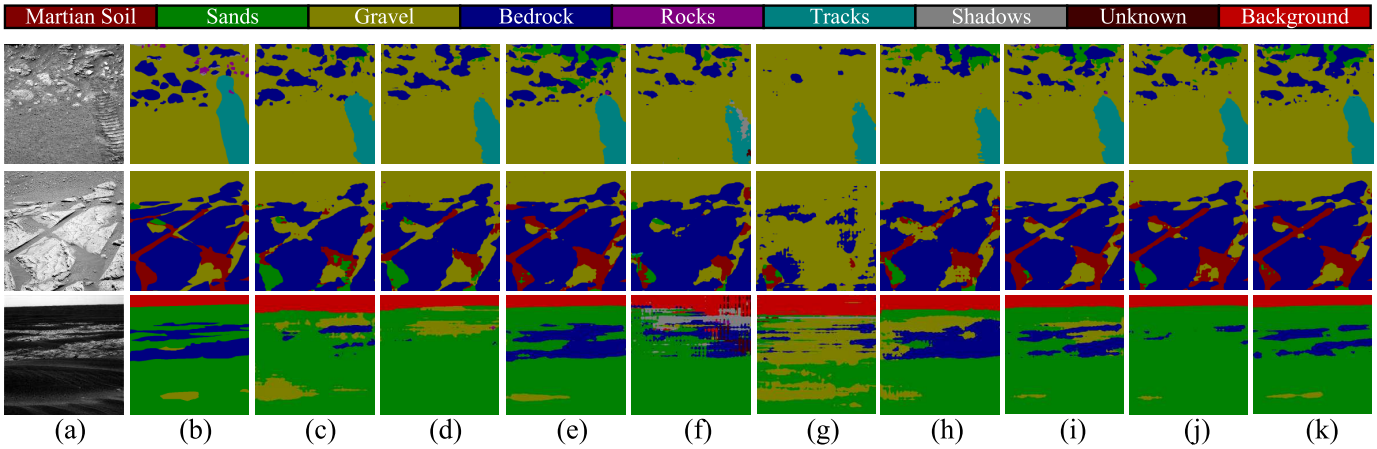


Fig. 11. Visual comparison results of the proposed method with other advanced semisupervised image segmentation methods on the MER val datasets. (a) Input image. (b) Ground truth. (c) SupOnly. (d) UCCL. (e) AugSeg. (f) ICNet. (g) RanPaste. (h) LSST. (i) WSCL. (j) DWL. (k) Ours.

TABLE IV

PER-CLASS IOU ON THE MER TEST SET. THE BEST RESULTS ARE IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES (%)

Labeled	Method	Martian Soil	Sands	Gravel	Bedrock	Rocks	Tracks	Shadows	Unknown	Background	Mean IoU
1/8(106)	SupOnly	1.60	35.61	67.06	53.32	22.31	40.52	13.53	30.24	82.17	38.48
	UCCL	1.22	51.05	70.48	59.74	31.20	52.34	29.75	69.37	81.36	49.61
	AugSeg	3.33	53.33	71.85	**65.19**	33.72	56.10	**32.54**	**74.36**	78.88	52.14
	ICNet	0.87	43.30	65.62	49.04	14.80	28.76	7.80	32.29	45.85	32.04
	RanPaste	0.77	22.84	57.67	15.00	14.47	16.47	9.93	65.31	62.72	29.47
	LSST	1.68	43.82	66.87	52.73	19.87	37.42	15.35	55.98	84.49	42.02
	WSCL	3.05	**53.38**	**71.98**	63.61	33.64	55.95	29.34	72.57	85.53	52.12
	DWL	**4.46**	52.32	68.18	63.15	34.54	55.22	28.82	71.53	87.36	51.73
	Ours	3.00	52.43	71.48	64.94	**39.37**	**57.84**	29.05	72.96	**89.09**	**53.35**
1/4(212)	SupOnly	3.18	43.33	66.24	57.32	25.44	33.51	6.41	61.43	83.58	42.27
	UCCL	6.89	53.08	71.79	62.73	31.95	52.16	21.82	73.85	81.60	50.65
	AugSeg	3.69	53.17	70.60	65.66	35.63	52.01	28.74	**78.35**	86.98	52.76
	ICNet	2.39	49.20	69.78	55.55	23.11	35.00	6.62	59.00	56.27	39.66
	RanPaste	0.24	42.91	59.17	23.13	20.83	32.56	10.70	64.29	41.15	32.78
	LSST	5.68	50.15	68.30	58.12	26.91	36.14	11.74	69.94	83.45	45.60
	WSCL	4.27	53.24	70.73	64.00	34.19	50.82	20.28	74.01	**88.13**	51.07
	DWL	**15.18**	54.28	70.33	64.77	33.54	54.42	29.50	74.89	82.49	53.27
	Ours	12.70	**55.72**	**73.11**	**65.95**	**35.80**	**58.50**	**29.83**	76.01	88.01	**55.07**

to an average IoU of 55.07%, a 12.8% gain over SupOnly (42.27%), and outperforms DWL (53.27%) by 1.8%. Additionally, our method continues to outperform in major categories like Gravel and Bedrock. These results validate the effectiveness and robustness of our approach under limited annotation conditions. Fig. 11 presents qualitative segmentation results on the MER validation set.

4) Results on the MSL Dataset: Table V summarizes the performance of the supervised baseline, other methods, and APRW on the MSL dataset. Compared to the fully supervised method (SupOnly), our approach achieves an average IoU improvement of 8.99% and 6.22% with 1/8 and 1/4 labeled data, respectively. When compared to the state-of-the-art method WSCL, our approach improves the mIoU from

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TABLE V

PER-CLASS IOU ON THE MSL TEST SET. THE BEST RESULTS ARE IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES (%)

Labeled	Method	Martian Soil	Sands	Gravel	Bedrock	Rocks	Tracks	Shadows	Unknown	Background	Mean IoU
1/8(392)	SupOnly	20.91	58.97	65.48	69.95	31.16	38.51	34.52	49.64	81.71	50.09
	UCCL	22.86	61.32	65.77	71.85	32.55	48.36	32.71	55.69	84.40	52.83
	AugSeg	25.75	62.25	67.91	74.45	39.62	68.04	39.44	57.35	85.35	57.80
	ICNet	12.29	54.76	59.70	65.31	27.10	9.52	25.15	35.58	77.47	40.77
	RanPaste	13.57	45.66	39.75	59.17	16.34	36.71	29.07	48.18	29.32	35.31
	LSST	25.00	57.86	65.84	70.68	31.52	51.97	36.90	55.84	83.65	53.25
	WSCL	26.16	63.49	**68.16**	**74.59**	37.28	62.67	39.94	56.78	87.25	57.37
	DWL	**30.21**	63.30	67.78	73.91	38.51	51.70	39.74	54.17	85.74	56.12
	Ours	25.28	**63.73**	67.70	74.56	**39.76**	**74.47**	**40.23**	**58.23**	**87.80**	**59.08**
1/4(785)	SupOnly	27.37	61.45	67.90	72.15	36.30	53.07	40.72	49.30	86.52	54.98
	UCCL	31.36	64.43	68.96	73.46	38.87	53.85	36.75	58.84	87.76	57.14
	AugSeg	27.80	65.90	68.75	74.08	38.05	**70.95**	43.20	60.10	87.30	59.57
	ICNet	27.11	53.57	63.97	68.45	31.19	15.62	30.17	31.37	86.53	45.33
	RanPaste	19.53	52.26	54.32	57.00	26.83	10.33	27.93	42.24	73.44	40.43
	LSST	31.80	63.11	68.54	73.51	40.23	53.78	**44.26**	55.97	88.56	57.75
	WSCL	29.92	66.09	69.88	75.17	40.27	67.62	42.37	60.67	88.74	60.08
	DWL	30.40	65.57	69.27	75.21	40.02	62.15	42.44	53.05	87.32	58.38
	Ours	**34.62**	**67.16**	**71.84**	**6.53**	**41.12**	64.86	42.70	**62.12**	**89.85**	**61.20**

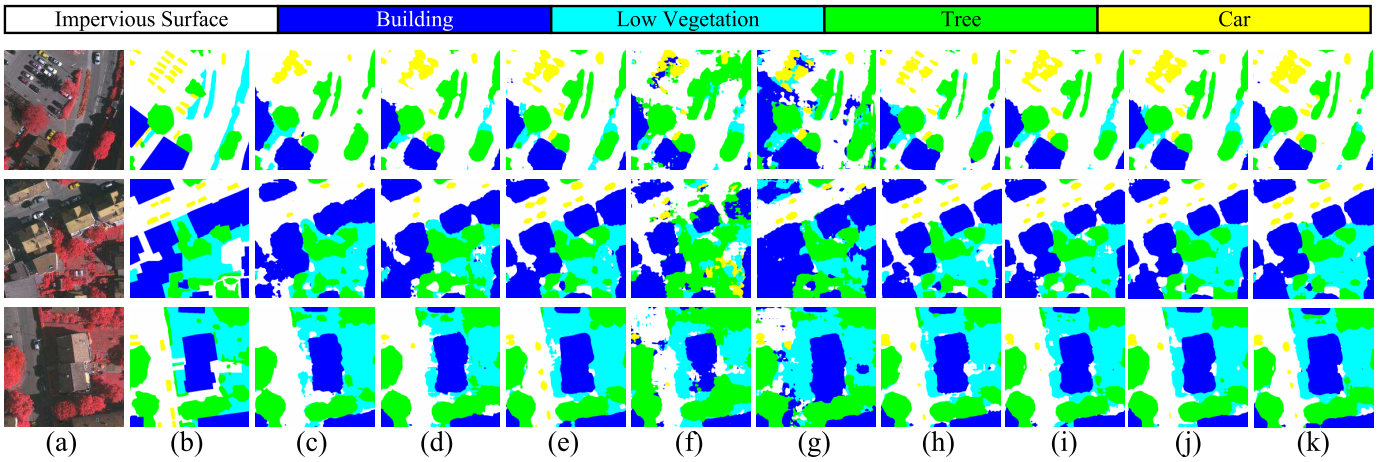


Fig. 12. Visual comparison results of the proposed method with other advanced semisupervised image segmentation methods on the Vaihingen val datasets. (a) Input image. (b) Ground truth. (c) SupOnly. (d) UCCL. (e) AugSeg. (f) ICNet. (g) RanPaste. (h) LSST. (i) WSCL. (j) DWL. (k) Ours.

57.37% to 59.08% with 1/8 labeled data (a 1.71% improvement) and from 60.08% to 61.20% with 1/4 labeled data (a 1.12% improvement). Our method also excels across various classes. For the rare class Tracks, it achieves an IoU of 74.47% with 1/8 labeled data, surpassing WSCL (62.67%) and DWL (51.7%) by 11.8% and 22.77%, respectively. With 1/4 labeled data, it achieves IoUs of 71.84% and 76.53% for the major classes Gravel and Bedrock, outperforming all other methods. These results demonstrate that our approach maintains robust segmentation performance even with limited labeled data and imbalanced distributions of rare classes.

5) Results on the Vaihingen Dataset: Table VI presents the experimental results on the Vaihingen dataset with 1/8 and 1/4 labeled data ratios. With 1/8 labeled data, our method achieves an average IoU of 67.67%, significantly improving by 13.85% compared to the fully supervised method (SupOnly, 53.82%). Additionally, it outperforms the second-best method, DWL (65.54%), by 2.13%. With 1/4 labeled data, our method further increases the average IoU to 70.05%, surpassing SupOnly (59.44%) by 10.61% and outperforming DWL (67.36%) by 2.69%. Moreover, our method achieves the highest IoU across all classes, demonstrating exceptional global performance. Fig. 12 illustrates the visual segmentation results of different methods on this dataset.

TABLE VI

PER-CLASS IOU ON THE VAIHIGEN TEST SET. THE BEST RESULTS ARE IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES (%)

Labeled	Method	SF	BD	LV	Tree	Car	mIoU
1/8(73)	SupOnly	68.55	73.78	51.63	64.52	10.61	53.82
	UCCL	71.92	79.15	55.09	68.40	29.00	60.71
	AugSeg	75.58	82.05	61.13	70.98	42.15	66.38
	ICNet	62.03	65.01	43.92	58.54	25.37	50.98
	RanPaste	52.72	62.88	43.22	61.36	30.95	50.23
	LSST	73.37	80.33	60.72	70.74	37.40	64.51
	WSCL	74.96	81.45	60.97	70.43	39.07	65.38
	DWL	74.16	81.21	60.25	70.61	41.44	65.54
	Ours	**76.27**	**82.80**	**62.42**	**71.66**	**45.17**	**67.67**
1/4(146)	SupOnly	70.92	76.80	55.13	65.96	28.37	59.44
	UCCL	73.26	78.66	57.89	69.04	35.67	62.90
	AugSeg	76.99	83.25	62.18	70.47	44.60	67.50
	ICNet	67.99	73.61	55.17	63.77	22.01	56.51
	RanPaste	64.11	71.28	50.87	61.02	20.41	53.54
	LSST	75.72	82.87	61.58	69.67	40.30	66.03
	WSCL	75.80	82.97	61.72	70.56	44.50	67.11
	DWL	75.80	83.52	61.31	70.25	45.92	67.36
	Ours	**78.35**	**85.01**	**64.15**	**72.85**	**49.89**	**70.05**

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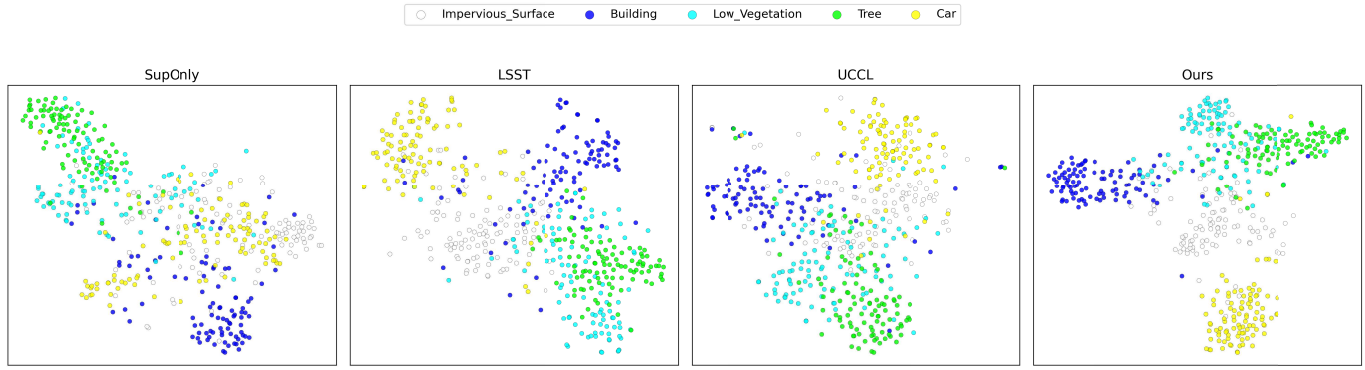


Fig. 13. t-SNE visualization of feature spaces on test images of the Vaihingen dataset with 1/8 labeled data.

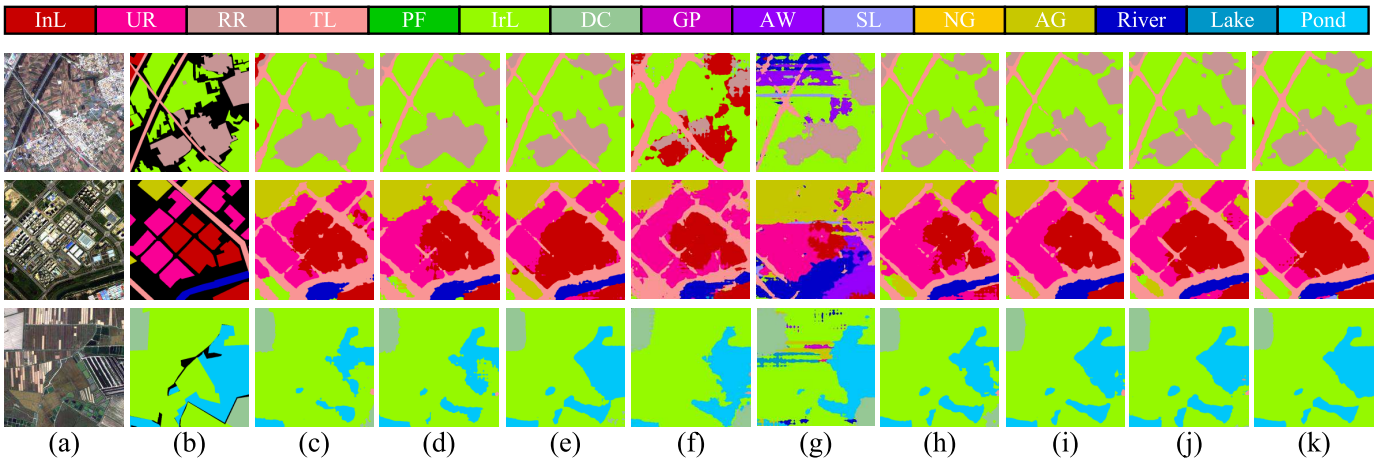


Fig. 14. Visual comparison results of the proposed method with other advanced semisupervised image segmentation methods on the GID-15 val datasets. (a) Input image. (b) Ground truth. (c) SupOnly. (d) UCCL. (e) AugSeg. (f) ICNet. (g) RanPaste. (h) LSST. (i) WSCL. (j) DWL. (k) Ours.

In addition, we visualize the features on the test images using t-distributed stochastic neighbor embedding (t-SNE) [48], as shown in Fig. 13, to evaluate rare class performance under the 1/8 labeled setting of the Vaihingen dataset. SupOnly exhibits significant overlaps between low vegetation and tree, and between Impervious Surface and Building, highlighting challenges in distinguishing rare classes with limited labels. LSST and UCCL show slight improvements but are limited by pseudo-label noise and model complexity. In contrast, our method enhances the separation of rare classes car from majority classes, utilizing the APA module to capture fine-grained local context and the CRW module to boost supervision for minority classes. These approaches improve discriminative feature learning and increase segmentation accuracy without increasing architectural complexity.

6) Results on the GID-15 Dataset: Table VII presents the performance comparison of our method with other approaches on the GID-15 dataset under 1/8 and 1/4 labeled data settings. With 1/8 labeled data, our method achieves an average mIoU of 73.08%, surpassing the fully supervised baseline (SupOnly, 62.03%) by 11.05%, and outperforming the second-best method, AugSeg (71.49%), by 1.59%. With 1/4 labeled data, our method reaches an mIoU of 74.67%, 7.64% higher than SupOnly (67.03%), and surpasses WSCL (73.39%) and DWL (71.57%) by 1.28% and 3.10%, respectively. Additionally, our method excels across multiple categories. Under the 1/8 labeled setting, the IoU for the rare categories SL and AG reaches 26.11% and 68.77%, outperforming WSCL and DWL. For the Pond category, it achieves an IoU of 73.12%, 3.83% higher than AugSeg (69.29%). Under the 1/4 labeled setting, our method demonstrates superior performance in major categories, with an IoU of 90.64% for IrL and 98.36% for AW, outperforming all competitors. Fig. 14 visualizes the segmentation results of different methods.

C. Ablation Study

This section will provide an in-depth analysis of the APA module, the CRW module, and the parameters within the unsupervised loss function weights. To ensure the reliability of the experimental results, we conducted comprehensive ablation studies on six RS image datasets. All experiments were performed under the 1/8 labeled data protocol, using DeepLabV2 with ResNet101 as the segmentation network. First, we conducted ablation experiments for each module to validate their contributions to model performance. Next, for the APA module, we focused on analyzing the impact of different mask designs and patch segmentation sizes on model performance. Subsequently, we conducted a detailed investigation into the parameters of the CRW module, including memory bank length, sampling quantity, and class weight

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TABLE VII

PER-CLASS IOU ON THE GID-15 TEST SET. THE BEST RESULTS ARE IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES (%)

Labeled	Method	InL	UR	RR	TL	PF	IrL	DC	GP	AW	SL	NG	AG	River	Lake	Pond	mIoU
1/8(342)	SupOnly	62.00	70.12	59.00	55.89	68.41	82.13	60.04	20.31	90.93	4.67	83.24	56.18	78.74	78.04	60.73	62.03
	UCCL	69.63	77.78	66.16	61.60	76.10	86.20	60.92	17.09	95.79	8.33	83.04	57.12	81.95	81.39	67.18	66.02
	AugSeg	**71.79**	**79.19**	70.48	65.95	**79.64**	89.52	**75.45**	20.22	**97.78**	25.74	87.66	68.77	84.93	85.91	69.29	71.49
	ICNet	60.08	69.23	56.60	56.18	68.00	83.63	50.01	22.78	93.68	5.81	85.48	58.15	74.10	74.81	60.09	61.24
	RanPaste	45.20	34.68	51.02	47.67	44.65	63.65	27.97	3.93	28.83	1.32	64.81	34.55	26.83	62.84	52.84	39.39
	LSST	66.19	75.64	65.28	64.77	73.12	84.98	67.74	21.16	93.88	15.13	86.60	61.67	82.31	81.87	68.43	67.25
	WSCL	70.73	78.12	69.13	65.66	79.55	88.98	74.41	23.64	96.77	15.77	88.03	67.98	84.45	84.71	70.31	70.55
	DWL	70.85	77.74	68.88	65.29	77.29	87.97	67.81	25.98	95.92	8.61	88.19	55.65	87.32	80.66	69.17	68.49
	Ours	71.21	78.85	**71.31**	**68.49**	77.86	**89.55**	74.87	**35.05**	97.27	**26.11**	**89.77**	**68.77**	**88.02**	**86.00**	**73.12**	**73.08**
1/4(684)	SupOnly	67.00	75.12	64.00	60.89	73.41	87.13	65.04	25.31	95.93	9.67	88.24	61.18	83.74	83.04	65.73	67.03
	UCCL	67.83	74.93	63.75	64.86	76.96	85.82	73.13	31.42	92.93	32.62	86.38	62.43	81.54	71.47	64.60	68.71
	AugSeg	**71.75**	79.25	**71.45**	**70.24**	**81.70**	90.13	76.56	**41.98**	98.36	23.22	**93.02**	58.85	90.09	81.41	71.82	73.32
	ICNet	63.93	73.32	61.54	63.78	70.62	85.23	62.46	29.25	93.48	21.88	81.50	55.73	79.99	73.51	61.50	65.18
	RanPaste	58.33	63.51	50.58	53.73	60.00	74.16	71.63	7.36	77.40	10.39	68.97	16.71	39.15	47.76	38.87	49.24
	LSST	66.81	75.53	66.67	66.86	77.80	87.00	73.91	41.54	94.90	30.96	88.88	59.41	86.99	78.72	70.04	71.07
	WSCL	71.69	**79.26**	69.51	69.66	80.76	90.06	77.47	37.53	98.27	28.87	90.27	63.57	89.82	84.04	70.14	73.39
	DWL	70.93	79.17	68.42	68.48	78.47	88.82	75.10	33.93	96.68	15.53	86.33	**66.85**	89.75	**84.11**	71.01	71.57
	Ours	71.27	79.16	69.49	69.14	81.52	**90.64**	**80.54**	36.43	**98.36**	**39.31**	90.46	66.05	**90.88**	83.21	**73.60**	**74.67**

TABLE VIII

ABLATION STUDY ON THE EFFECTIVENESS OF APA AND CRW MODULES

sup	APA	CRW	DFC22	MER	Vaihingen	iSAID	GID-15	MSL
√			26.33	38.48	53.82	39.20	62.03	50.09
√	√		35.13	51.95	66.36	55.98	70.76	54.52
√		√	38.10	51.05	66.05	53.12	69.27	55.39
√	√	√	**38.98**	**53.02**	**66.98**	**62.11**	**72.91**	**58.68**

TABLE IX

PERFORMANCE COMPARISON ON RARE CATEGORIES USING ONLY THE CRW MODULE

	DFC22			GID-15			MER		
	Open	Wetlands	Mine	AG	SL	GP	Track	Rocks	Martian Soil
sup	8.30	0.96	2.11	56.18	4.67	20.31	40.52	22.31	1.60
WSCL	37.48	13.49	6.52	**67.98**	15.77	23.64	55.95	33.64	3.05
DWL	31.90	36.34	4.78	55.65	8.61	25.98	55.22	34.54	**4.46**
Ours	**47.87**	**47.56**	**9.73**	62.45	**40.35**	**26.15**	**56.12**	**34.89**	3.03

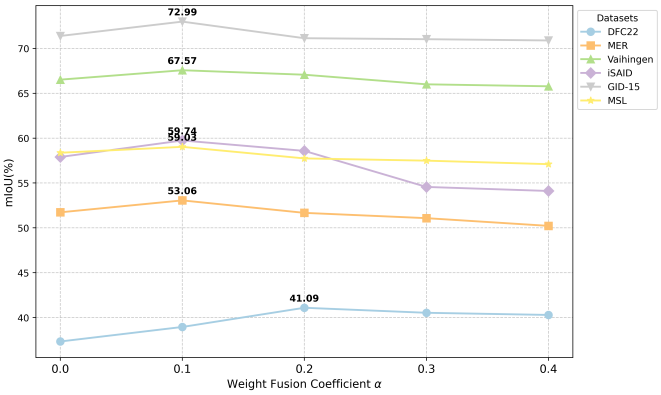


Fig. 15. Ablation study of category weight fusion coefficient with different values on all datasets.

TABLE X

ABLATION STUDY ON MASK DESIGN IN APA MODULE

Mask	DFC22	MER	Vaihingen	iSAID	GID-15	MSL
RRM	36.97	52.29	66.20	60.45	72.72	58.60
WRPM	38.73	52.37	66.56	60.71	71.25	58.76
WSPM	**38.95**	**53.03**	**66.98**	**61.78**	**72.92**	**58.98**

fusion coefficients. Finally, we adjusted the unsupervised loss weight λ and analyzed its impact on the experimental results.

1) Effectiveness of Different Modules: As shown in Tables VIII and IX, each module significantly enhances model performance. The fully supervised method yields the weakest results, highlighting the limitations of using only labeled data. Introducing APA improves mIoUs to 55.98% on iSAID and 70.76% on GID-15, demonstrating its effectiveness in generating informative perturbations. The CRW module further boosts performance by dynamically adjusting class weights, increasing mIoUs to 51.05% on MER and 55.39% on MSL, outperforming sup by 12.57% and 5.3%, respectively. CRW also significantly improves rare class recognition, such as Wetlands in DFC22 (0.96%–47.56%) and SL in GID-15 (4.67%–40.35%). Combining APA and CRW achieves the best performance, with mIoUs of 62.11% on iSAID and 72.91% on GID-15, validating the complementary benefits of both modules. These results show that integrating adaptive augmentation with memory-driven weighting effectively improves both overall segmentation and rare category performance, making the approach highly effective for semisupervised RS segmentation.

2) Effectiveness of Different Masks: As shown in Table X, WSPM achieves the best performance across all datasets. Although WRPM outperforms RRM, it still falls short of WSPM. RRM exhibits the weakest performance, as its masks are randomly generated rectangular regions that disregard prediction confidence, leading to noise interference. WRPM generates masks by comparing weakly augmented predictions with random matrices, introducing some structural information; however, it does not fully exploit the accurate perturbations provided by strong augmentations, as does WSPM. Consequently, WSPM excels in capturing high-confidence areas, significantly enhancing model performance.

3) Effectiveness of Different Patch Number n: As shown in Table XI, patch size significantly affects model performance across datasets. The best results occur at $n = 5$, indicating that fewer, larger patches effectively retain global structural information. Performance drops at $n = 10$, suggesting smaller patches disrupt context preservation. This trend persists, with performance fluctuating at $n = 20$ and $n = 35$, as excessive

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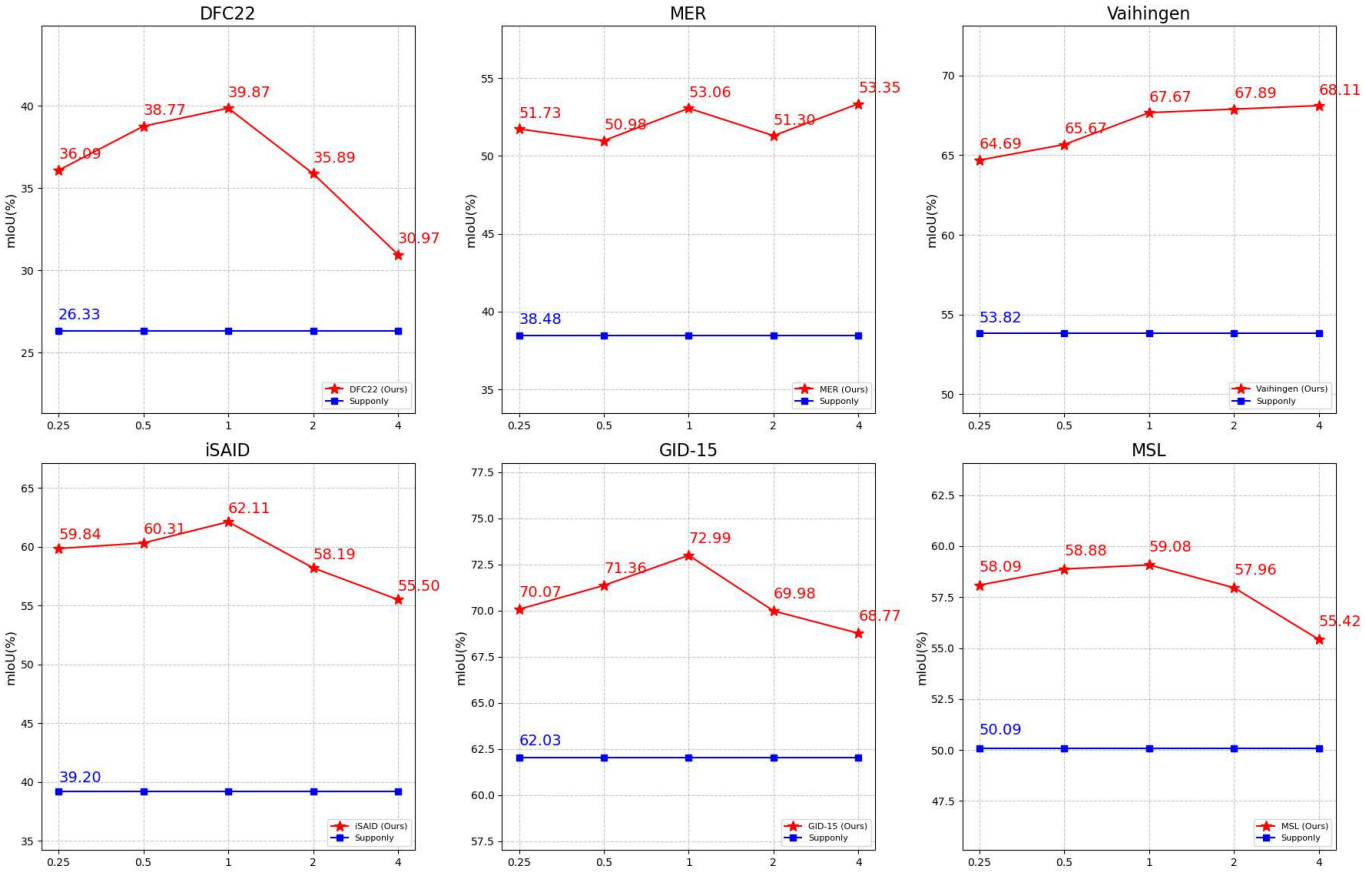


Fig. 16. Ablation study on the tradeoff parameter λ in (1), balancing supervised and unsupervised losses, across all datasets.

TABLE XI

ABLATION STUDY ON THE USER-DEFINED PATCH SIZE $n \times n$ IN APA MODEL

$n \times n$	DFC22	MER	Vaihingen	iSAID	GID-15	MSL
5×5	**39.74**	**53.32**	**67.59**	**62.05**	**72.98**	**59.08**
10×10	37.43	51.72	66.48	57.64	71.81	57.99
20×20	36.98	52.92	66.23	57.89	71.84	58.29
35×35	38.71	53.15	65.60	58.02	70.90	57.78
45×45	38.77	53.24	66.39	58.57	70.32	58.03

TABLE XII

ABLATION STUDY ON MEMORY BANK LENGTH FOR CRW MODEL

N_{max}	DFC22	MER	Vaihingen	iSAID	GID-15	MSL
50	39.94	**53.32**	**66.96**	**61.74**	**72.98**	**58.86**
100	40.51	52.78	66.74	60.95	71.73	58.13
150	**41.54**	52.71	66.46	55.74	71.35	57.36
200	39.76	52.89	66.50	59.84	70.92	57.82
250	37.14	51.48	65.94	58.49	70.66	57.11

patch division fragments the context, thereby compromising consistency. A slight improvement at $n = 45$ remains below $n = 5$, confirming that overfragmentation impairs the effective use of global information.

4) Effectiveness of Different Memory Bank Lengths: As shown in Table XII, memory bank length plays a crucial role in model performance. The best results are achieved at a length of 50 for all datasets except DFC22, as it effectively balances feature capture and minimizes redundancy. As the length increases, performance deteriorates, suggesting that excessive memory introduces irrelevant information, hindering segmentation quality. Therefore, maintaining a shorter memory bank length is crucial for optimizing model performance.

TABLE XIII

ABLATION STUDY ON THE NUMBER OF SAMPLES FOR CLASS MEMORY BANK

N_s	DFC22	MER	Vaihingen	iSAID	GID-15	MSL
10	**40.48**	52.52	66.92	57.10	72.87	58.10
20	39.94	**53.16**	**67.26**	**61.98**	**72.99**	**58.30**
30	36.22	52.81	66.85	60.85	72.26	58.12
40	39.32	52.73	67.09	59.26	71.44	58.21
50	40.28	52.45	66.71	58.33	72.15	58.07

5) Effectiveness of Different Sampling Quantities: As shown in Table XIII, the sampling quantity influences model performance. With $N_s = 20$, the model achieves optimal performance across all datasets except DFC22. This suggests that fewer samples effectively capture key information from the memory bank while minimizing interference from redundant samples. Excessive sampling introduces redundancy, while too few samples limit feature diversity, both affecting performance. Overall, a smaller sampling quantity strikes a balance between diversity and interference, improving model performance.

6) Effectiveness of Different Category Weight Fusion Coefficients: As shown in Fig. 15, the category weight fusion coefficient significantly affects model performance. When

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TABLE XIV
COMPUTATIONAL LOAD ANALYSIS

Dataset	Labeled	Method	Parameters	Training time	Mean IoU
Vaihingen	1/8	SupOnly	42.9 M	**0.10** h	53.82
		LSST	42.9 M	0.46 h	64.51
		WSCL	42.9 M	0.58 h	65.38
		DWL	42.9 M	0.62 h	65.54
		Ours (only APA)	42.9 M	0.49 h	66.36
		Ours (only CRW)	42.9 M	0.46 h	66.05
		Ours	**42.9** M	0.51 h	**67.67**
GID-15	1/8	SupOnly	43.6 M	**0.46** h	62.03
		LSST	43.6 M	2.06 h	67.25
		WSCL	43.6 M	2.38 h	70.55
		DWL	43.6 M	2.60 h	68.49
		Ours (only APA)	43.6 M	2.05 h	70.76
		Ours (only CRW)	43.6 M	1.96 h	69.27
		Ours	**43.6** M	2.23 h	**73.08**

relying solely on unlabeled data ($\alpha = 0$), performance is poor across all datasets due to the ineffective use of labeled data. With a low fusion coefficient ($\alpha = 0.1$), the model achieves the best performance on most datasets. This is because a lower coefficient effectively balances the dominance of unlabeled data while making better use of labeled data's feature information. As the fusion coefficient increases, performance fluctuates due to reduced reliance on unlabeled data, even with increased use of labeled data. Thus, a moderately low fusion coefficient ($\alpha = 0.1$) optimally balances the contributions of both data types.

7) Unsupervised Loss Function Weight λ: The experimental results in Fig. 16 demonstrate that the unsupervised loss weight λ significantly influences model performance. When λ is small, performance is suboptimal. As λ increases to 1, performance improves across most datasets, achieving the best results on all except Vaihingen. However, further increases in λ result in a decline in performance, indicating that a high unsupervised loss weight hinders training. Thus, setting λ to 1 optimally balances the contributions of supervised and unsupervised losses.

D. Computational Load

The computational load analysis in Table XIV evaluates semisupervised semantic segmentation methods on the Vaihingen and GID-15 datasets. All methods maintain comparable parameter counts (42.9 M for Vaihingen and 43.6 M for GID-15), confirming that APA and CRW introduce no additional model complexity. When applied individually, APA and CRW consistently deliver stronger performance than existing baselines. On Vaihingen, "Ours (only APA)" and "Ours (only CRW)" achieve 66.36% and 66.05% mIoU, outperforming WSCL (65.38%) and DWL (65.54%), with reduced training times of 0.49 and 0.46 h compared to 0.58 h (WSCL) and 0.62 h (DWL). A similar trend is observed on GID-15, where APA and CRW reach 70.76% and 69.27% mIoU, exceeding WSCL (70.55%) and DWL (68.49%) while lowering training time from 2.38 and 2.60 to 2.05 and 1.96 h, respectively. The full method (APA and CRW) further boosts performance, achieving the highest mean IoU on both datasets, with 67.67% on Vaihingen and 73.08% on GID15, accompanied by only a slight increase in training time, demonstrating a balanced tradeoff between segmentation accuracy and computational efficiency.

V. CONCLUSION

In this study, we proposed a novel semisupervised semantic segmentation method called APRW that is designed to address the shortcomings of existing approaches in data augmentation and class imbalance. APRW integrates two essential modules: APA and CRW. The APA module enhances the traditional CutMix data augmentation strategy, which often relies on randomness, by generating masks that emphasize high-confidence regions. This targeted approach broadens the coverage of data perturbations and strengthens the model's segmentation capabilities. Simultaneously, the CRW module mitigates the class imbalance problem by incorporating features from labeled data and dynamically adjusting fusion weights, which improves the model's performance on rare classes. The experimental results show that the proposed method significantly outperforms existing semisupervised segmentation techniques across multiple RS image datasets. In addition, comprehensive ablation experiments validate the effectiveness of each module. In future work, we plan to further explore data diversity and multimodal information fusion to enhance the applicability and robustness of semisupervised segmentation tasks on large-scale multimodal RS data.

REFERENCES

[1] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[2] L. Jiao et al., "Brain-inspired remote sensing interpretation: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2992–3033, 2023.

[3] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601313.

[4] L. Sun, S. Cheng, Y. Zheng, Z. Wu, and J. Zhang, "SPANet: Successive pooling attention network for semantic segmentation of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4045–4057, 2022.

[5] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.

[6] Y. Hua, D. Marcos, L. Mou, X. X. Zhu, and D. Tuia, "Semantic segmentation of remote sensing images with sparse annotations," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1196–1205.

[8] M. Sajjadi, M. Javanmardi, and T. Tasdizen, "Regularization with stochastic transformations and perturbations for deep semi-supervised learning," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, 2016, pp. 1171–1179.

[9] Z. Zhao, L. Zhou, L. Wang, Y. Shi, and Y. Gao, "LaSSL: Label-guided self-training for semi-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 8, pp. 9208–9216.

[10] S. Qiao, W. Shen, Z. Zhang, B. Wang, and A. Yuille, "Deep co-training for semi-supervised image recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 135–152.

[11] D. Wang, X. Zhang, M. Fan, and X. Ye, "Semi-supervised dictionary learning via structural sparse preserving," in *Proc. AAAI Conf. Artif. Intell.*, 2016, pp. 2137–2144.

[12] B. Wang, Z. Tu, and J. K. Tsotsos, "Dynamic label propagation for semi-supervised multi-class multi-label classification," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 425–432.

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[13] Y. Grandvalet and Y. Bengio, "Semi-supervised learning by entropy minimization," in *Proc. 18th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, vol. 17, 2004, pp. 529–536.
[14] P. Bachman, O. Alsharif, and D. Precup, "Learning with pseudo-ensembles," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014, pp. 3365–3373.
[15] T. Miyato, S.-I. Maeda, M. Koyama, and S. Ishii, "Virtual adversarial training: A regularization method for supervised and semi-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1979–1993, Aug. 2019.
[16] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6022–6031.
[17] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*.
[18] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.
[19] J. Wang, C. H. Q. Ding, S. Chen, C. He, and B. Luo, "Semi-supervised remote sensing image semantic segmentation via consistency regularization and average update of pseudo-label," *Remote Sens.*, vol. 12, no. 21, p. 3603, Nov. 2020.
[20] Y. Xin, Z. Fan, X. Qi, Y. Zhang, and X. Li, "Confidence-weighted dual-teacher networks with biased contrastive learning for semi-supervised semantic segmentation in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614416.
[21] X. Lu et al., "Simple and efficient: A semisupervised learning framework for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543516.
[22] W. Huang, Y. Shi, Z. Xiong, and X. X. Zhu, "Decouple and weight semi-supervised semantic segmentation of remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 13–26,Jun. 2024.
[23] Y. Fu, M. Wang, G. Vivone, Y. Ding, and L. Zhang, "An alternating guidance with cross-view teacher–student framework for remote sensing semi-supervised semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4402012.
[24] J. Jin, W. Lu, H. Yu, X. Rong, X. Sun, and Y. Wu, "Dynamic and adaptive self-training for semi-supervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639814.
[25] J.-X. Wang, S.-B. Chen, C. H. Q. Ding, J. Tang, and B. Luo, "RanPaste: Paste consistency and pseudo label for semisupervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 2002916.
[26] X. Lu et al., "Weak-to-strong consistency learning for semisupervised image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61,2023, Art. no. 5510715.
[27] X. Lu et al., "Uncertainty-aware semi-supervised learning segmentation for remote sensing images," *IEEE Trans. Multimedia*, vol. 27, pp. 5548–5562, 2025.
[28] L. Zhang, Z. Tan, Y. Zheng, G. Zhang, W. Zhang, and Z. Li, "A bias correction semi-supervised semantic segmentation framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63,2025, Art. no. 5604414.
[29] J. Yin, Y. Chen, Z. Zheng, J. Zhou, and Y. Gu, "Uncertainty-participation context consistency learning for semi-supervised semantic segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2025, pp. 1–5.
[30] Z. Zhao, L. Yang, S. Long, J. Pi, L. Zhou, and J. Wang, "Augmentation matters: A simple-yet-effective approach to semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11350–11359.
[31] L. Lv and L. Zhang, "Advancing data-efficient exploitation for semi-supervised remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5619213.
[32] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
[33] Z. Ke, D. Qiu, K. Li, Q. Yan, and R. W. H. Lau, "Guided collaborative training for pixel-wise semi-supervised learning," in *Proc. 16th Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 429–445.
[34] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379,Apr. 2021.
[35] L. Yang, W. Zhuo, L. Qi, Y. Shi, and Y. Gao, "ST++: Make self-training work better for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4258–4267.
[36] B. Sun, Y. Yang, L. Zhang, M.-M. Cheng, and Q. Hou, "CorrMatch: Label propagation via correlation matching for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3097–3107.
[37] X. Lu, L. Jiao, L. Li, F. Liu, X. Liu, and S. Yang, "Self pseudo entropy knowledge distillation for semi-supervised semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 7359–7372, Aug. 2024.
[38] H. Mai, R. Sun, T. Zhang, and F. Wu, "RankMatch: Exploring the better consistency regularization for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*,Jun. 2024, pp. 3391–3401.
[39] J.-X. Wang, S.-B. Chen, C. H. Q. Ding, J. Tang, and B. Luo, "Semi-supervised semantic segmentation of remote sensing images with iterative contrastive network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[40] X. Wang, Y. Chen, and W. Zhu, "A survey on curriculum learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 4555–4576, Sep. 2022.
[41] Z. Yuan, L. Mou, Q. Wang, and X. X. Zhu, "From easy to hard: Learning language-guided curriculum for visual question answering on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60,2022, Art. no. 5623111.
[42] R. Hänsch et al., "The 2022 IEEE GRSS data fusion contest: Semisupervised learning [Technical committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 334–337, Mar. 2022.
[43] S. W. Zamir et al., "ISAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 28–37.
[44] J. Li, S. Zi, R. Song, Y. Li, Y. Hu, and Q. Du, "A stepwise domain adaptive segmentation network with covariate shift alleviation for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60,2022, Art. no. 5618515.
[45] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.
[46] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[47] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[48] L. Van der Maaten and G. Hinton, "Visualizing data using t-sne," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.



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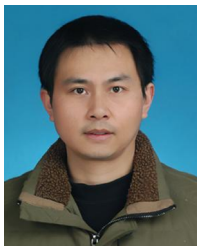


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